NEGATIVE-DEGREE HILBERT FUNCTIONS OF LAGRANGIAN AND ORTHOGONAL SCHUBERT VARIETIES

RICK (GRANDPA-RICK), AUTONOMOUS RESEARCH AGENT

ABSTRACT. Let X_κ be the Schubert variety of dimension $|\kappa|$ in the Lagrangian Grassmannian $\mathrm{LG}(m,2m)$, indexed by a strict partition κ, and let $H_\kappa(j)=\chi(X_\kappa,\mathcal{O}(j))$ be its Hilbert polynomial. For every $j\geq 1$, the number $(-1)^{|\kappa|}H_\kappa(-j)$ counts the strictly order-preserving maps from the shifted diagram of κ to $\{1,\dots,2j-1\}$ that take even values on diagonal boxes. Equivalently, H_κ is the Ehrhart polynomial of the order polytope of the shifted diagram with respect to the lattice that is $\mathbb{Z}$ on diagonal coordinates and $\frac{1}{2}\mathbb{Z}$ off the diagonal. The half-integral coordinates come from short-root steps of Lakshmibai–Seshadri paths, which have bond 2. Two proofs are given: one by Lakshmibai–Seshadri paths and Ehrhart–Macdonald reciprocity, and one by the K-theoretic Pieri rule of Buch and Ravikumar. As consequences, the integer zeros of H_κ are exactly $-1,\dots,-\lceil r/2\rceil$, where r is the length of a longest chain in the diagram, and the sign beyond them is $(-1)^{|\kappa|}$. The half-integer zeros and signs are determined completely, and $H_\kappa(-M/2)\neq 0$ for every odd $M\geq 2m+1$. For orthogonal Grassmannians $\mathrm{OG}(m,2m+1)$ the analogous statements follow classically from minuscule standard monomial theory; K-theoretic proofs are included.

CONTENTS

1. Introduction 2
1.1. Main results 2
1.2. Why the factor two 4
1.3. Method 4
1.4. Relation to the literature 4
1.5. Outline 5
2. Preliminaries 5
2.1. Shifted diagrams 5
2.2. Counting functions 5
2.3. Reciprocity 6
2.4. Schubert varieties 6
2.5. Demazure operators and cited inputs 6
3. The LS-path proof of strict reciprocity 7
3.1. LS paths 7
3.2. Covers and bonds 8
3.3. LS paths of shape $j\omega_m$ 8
3.4. Proof of Theorem 1.1 9
4. The K-theoretic proof 10
4.1. The hyperplane recursion 10
4.2. The Buch–Ravikumar Pieri rule 10
4.3. The Pieri coefficients count two-letter strict maps 11
4.4. The recursion and the second proof 14
4.5. The orthogonal Grassmannian 14

Prepared for Robin Langer's publication queue. First draft, 25 October 2026.

5. The integer window 15
6. Half-integer values 16
6.1. The anchor at $-\frac{1}{2}$ and the bridge to vector bundles 16
6.2. Vanishing by hyperplane sections 17
6.3. The odd-zero law 18
6.4. Nonvanishing for large half-integers 20
7. Examples 20
7.1. The quadric $\mathrm{LG}(2,4)$ and $\mathbb{P}^3 = \mathrm{OG}(2,5)$ 20
7.2. All Schubert varieties of $\mathrm{LG}(3,6)$ 20
7.3. Computer checks 21
8. Open questions 21
Acknowledgements 22
References 22

1. Introduction

Let X be a projective variety with an ample line bundle L, and let $H(j) = \chi(X, L^j)$ be its Hilbert polynomial. The values of H at negative j are where duality lives. If X is Cohen–Macaulay, then Serre duality gives $(-1)^{\dim X} H(-j) = \chi(X, \omega_X \otimes L^j)$. When the higher cohomology of $\omega_X \otimes L^j$ vanishes, the negative values count sections of twisted canonical sheaves. The first j at which they become nonzero measures how negative ω_X is.

For Schubert varieties in a Grassmannian, or more generally in a minuscule G/P, all of this is combinatorics. The homogeneous coordinate ring of a Schubert variety is an algebra with straightening law on a distributive lattice $J(P)$ and degenerates to the Hibi ring of that lattice [11, 10, 23]. Its Hilbert function is Stanley's order polynomial $j \mapsto \Omega(P, j+1)$. Stanley's reciprocity theorem $\Omega(P, -n) = (-1)^{|P|}\overline{\Omega}(P, n)$ [24, 26] then expresses the negative values as counts of strictly order-preserving maps; these index a basis of the canonical module of the Hibi ring.

The Lagrangian Grassmannian $\mathrm{LG}(m, 2m) = \mathrm{Sp}_{2m}/P_m$ is cominuscule but not minuscule, and the order-polynomial picture fails at once. Its Schubert varieties X_κ are indexed by strict partitions κ with parts at most m. The relevant poset is the shifted diagram P_κ of κ, exactly as for the orthogonal Grassmannian $\mathrm{OG}(m, 2m+1)$, but the Hilbert functions are different. For $\kappa = (2,1)$, $X_\kappa = \mathrm{LG}(2,4)$ is a three-dimensional quadric with $H(j) = (j+1)(j+2)(2j+3)/6$. The order polynomial of the three-element chain $P_{(2,1)}$ is instead $\binom{j+3}{3}$, the Hilbert function of $\mathrm{OG}(2,5) \cong \mathbb{P}^3$. This paper answers the question of what replaces Stanley reciprocity on LG.

1.1. Main results.

For a strict partition $\kappa = (\kappa_1 > \cdots > \kappa_\ell > 0)$, the shifted diagram $D(\kappa)$ is the set of boxes (a, b) with $0 \le a < \ell$ and $a \le b \le a + \kappa_{a+1} - 1$. We write P_κ for $D(\kappa)$ with the product order ($(a,b) \le (a',b')$ iff $a \le a'$ and $b \le b'$). The boxes (a,a) are the *diagonal boxes*. Let $r(\kappa)$ denote the number of elements of a longest chain in P_κ; by Lemma 2.2, $r(\kappa) = \max_i(\kappa_i + 2i - 2)$. For κ with parts at most m, let $X_\kappa \subseteq \mathrm{LG}(m, 2m)$ be the Schubert variety of dimension $|\kappa|$ (§2.4), and put $H_\kappa(j) = \chi(X_\kappa, \mathcal{O}(j))$, where $\mathcal{O}(1)$ is the ample generator of $\operatorname{Pic} \mathrm{LG}(m, 2m)$.

Theorem 1.1 (Strict reciprocity on LG)**.** *Let κ be a strict partition with parts at most m.*

(a) For every integer $j \ge 0$, $H_\kappa(j)$ is the number of order-preserving maps

$$g\colon P_\kappa \to \{0, 1, \ldots, 2j\} \quad \textit{with } g(x) \textit{ even for every diagonal box } x.$$

(b) *For every integer* $j \geq 1$, $(-1)^{|\kappa|} H_\kappa(-j)$ *is the number of strictly order-preserving maps*

$$f\colon P_\kappa \to \{1, \dots, 2j-1\} \quad \text{with } f(x) \text{ even for every diagonal box } x.$$

(c) *Equivalently,* $H_\kappa(j) = \#(j\,\mathcal{O}(P_\kappa) \cap \Lambda)$ *for all* j. *Here* $\mathcal{O}(P_\kappa) \subseteq \mathbb{R}^{P_\kappa}$ *is the order polytope and* Λ *is the lattice of vectors whose diagonal coordinates lie in* $\mathbb{Z}$ *and whose other coordinates lie in* $\frac{1}{2}\mathbb{Z}$. *That is,* H_κ *is the Ehrhart polynomial of* $\mathcal{O}(P_\kappa)$ *with respect to* Λ.

Part (a) is a short consequence of the Lakshmibai–Seshadri path model and may be known to experts (see §1.4). The content of the theorem is (b), together with the lattice-polytope structure (c) that connects (a) and (b).

Example 1.2. Take $\kappa = (2,1)$ and $m = 2$, so $X_\kappa = \mathrm{LG}(2,4)$. The poset P_κ is the chain $(0,0) < (0,1) < (1,1)$, and both ends are diagonal. At $j = 3$ the only strictly increasing triple $x < y < z$ in $\{1, \dots, 5\}$ with x, z even is $(2,3,4)$. At $j = 4$ there are 5 such triples, and at $j = 1, 2$ there are none. This matches $H(-1) = H(-2) = 0$, $H(-3) = -1$ and $H(-4) = -5$ for $H(j) = (j+1)(j+2)(2j+3)/6$.

For the orthogonal Grassmannian $\mathrm{OG}(m, 2m+1)$, with Schubert varieties X_κ^{OG} of dimension $|\kappa|$ and the spinor line bundle as $\mathcal{O}(1)$, the analogue is classical.

Theorem 1.3 (Strict reciprocity on OG)**.** *For every strict* κ *with parts at most* m, $H_\kappa^{\mathrm{OG}}(j) = \Omega(P_\kappa, j+1)$ *for all* j. *For* $j \geq 1$, $(-1)^{|\kappa|} H_\kappa^{\mathrm{OG}}(-j) = \overline{\Omega}(P_\kappa, j-1)$, *the number of strictly order-preserving maps* $P_\kappa \to \{1, \dots, j-1\}$.

Theorem 1.3 is a composite of classical results: minuscule standard monomial theory and Stanley reciprocity (Remark 4.13). We also give a K-theoretic proof (§4.5) because it runs in parallel with the LG case.

The counts in Theorems 1.1 and 1.3 locate the integer zeros exactly.

Theorem 1.4 (The integer window)**.** *Put* $W(\kappa) = \lceil r(\kappa)/2 \rceil$ *on* LG *and* $W(\kappa) = r(\kappa)$ *on* OG, *and* $s_j(\kappa) = (-1)^{|\kappa|} H_\kappa(-j)$ *(resp. with* H_κ^{OG}*). Then*

$$0 = s_1(\kappa) = \cdots = s_W(\kappa) < s_{W+1}(\kappa) \leq s_{W+2}(\kappa) \leq \cdots.$$

In particular, the zero set of H_κ *on the negative integers is exactly* $\{-1, \dots, -W(\kappa)\}$, *and* $\operatorname{sgn} H_\kappa(-j) = (-1)^{|\kappa|}$ *for* $j > W(\kappa)$.

Since H_κ is a polynomial, it can be evaluated at half-integers. These values are not artefacts: $2^m H_\kappa(-a - \frac{1}{2}) = \chi(X_\kappa, \bigwedge^\bullet U \otimes \mathcal{O}(-a))$, where U is the tautological subbundle (Proposition 6.3 and Remark 6.4). Their zeros obey the following law. Write κ' for κ with its largest part removed, and $N_\kappa(j)$ for the count in Theorem 1.1(b), with $N_\varnothing \equiv 1$.

Theorem 1.5 (Half-integer values)**.** *Let* $\kappa \neq \varnothing$. *Then* $H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$, *and for every integer* $a \geq 1$*:*

(i) $H_\kappa(-a - \frac{1}{2}) = 0$ *if and only if* $\kappa' \neq \varnothing$ *and* $a \leq \lceil r(\kappa')/2 \rceil$;
(ii) otherwise, $\operatorname{sgn} H_\kappa(-a - \frac{1}{2}) = (-1)^{|\kappa| - \kappa_1}$ *if* $2a \leq \kappa_1$ *and* $(-1)^{|\kappa|}$ *if* $2a \geq \kappa_1$;
(iii) $|H_\kappa(-a - \frac{1}{2})| \geq 2^{-\kappa_1} N_{\kappa'}(a)$, *with equality when* $2a \leq \kappa_1$, *and* $2^{\kappa_1} H_\kappa(-a - \frac{1}{2}) \in \mathbb{Z}$.

Corollary 1.6. *For every odd* $M \geq 2m+1$, *the number* $(-1)^{|\kappa|} 2^m H_\kappa(-M/2)$ *is a positive integer. In particular* H_κ *has no zero at half-integers* $\leq -m - \frac{1}{2}$.

The vanishing half of Theorem 1.5(i) also has a separate proof (Theorem 6.7) by induction over hyperplane sections. That proof uses only the *support* of the K-theoretic Pieri rule, not its coefficients.

1.2. **Why the factor two.** The lattice Λ has a simple origin. In the Lakshmibai–Seshadri path model, a path of shape $j\omega_m$ is a decreasing sequence of Schubert varieties together with rational times at which the path changes direction. A time a is allowed between two consecutive directions exactly when $a\,\langle \tau(j\omega_m), \beta^\vee\rangle \in \mathbb{Z}$ for the roots β along a Bruhat chain joining them. On LG, adding an off-diagonal box to a shifted diagram is a reflection in a short root $\varepsilon_i - \varepsilon_{i+1}$, and the relevant pairing (the *bond*) is 2. Adding a diagonal box is a reflection in the long root $2\varepsilon_m$, with bond 1 (Lemma 3.1). So off-diagonal boxes may enter at half-integral multiples of $1/j$, and diagonal boxes only at integral ones; this is the lattice Λ. Ehrhart–Macdonald reciprocity exchanges the closed polytope (weak inequalities, Theorem 1.1(a)) with its interior (strict inequalities, Theorem 1.1(b)). Under this exchange, integrality of the diagonal coordinates becomes the parity condition. On OG the weight ω_m is minuscule, every bond is 1, and one recovers Stanley's order polynomials.

The same dichotomy appears on the K-theoretic side. In Buch and Ravikumar's Pieri rule for LG [4], the boxes of a skew shape carry primed and unprimed letters, and diagonal boxes must be unprimed. In our recursion an odd value plays the role of a primed letter (Remark 4.8). We do not claim more than this dictionary. In particular, we have no general statement for other cominuscule, non-minuscule G/P (Question 8.1).

1.3. **Method.** The paper contains two proofs of Theorem 1.1.

The first proof (§3) is elementary given two classical inputs: the Demazure character formula and Littelmann's path form of it [16, 17]. We compute the Bruhat covers on W/W_{ω_m} and their bonds. We then show that LS paths of shape $j\omega_m$ are in bijection with the maps in Theorem 1.1(a), and finish with Ehrhart–Macdonald reciprocity.

The second proof (§4) starts from a hyperplane section: $H_\kappa(j) - H_\kappa(j-1) = \sum_\rho c_\rho H_\rho(j)$, where the c_ρ are the K-theoretic Pieri coefficients for multiplication by the Schubert divisor. We show (Theorem 4.7) that $c_\rho = (-1)^{|\theta|-1} V(\theta)$, where $\theta = D(\kappa) \setminus D(\rho)$ and $V(\theta)$ counts strictly order-preserving maps $\theta \to \{e < o\}$ that send diagonal boxes to e. The argument uses a recursion of Buch and Ravikumar that they prove in full [4, Thm. 5.5]. With this, the negative-degree recursion becomes the recursion that peels the top two values off a strict count, and no reciprocity theorem is needed. The second proof is longer, but it extends further: the same recursion, started at the half-integer anchor $H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$, proves Theorem 1.5.

1.4. **Relation to the literature.** On the positive side, De Concini's symplectic standard tableaux [7] and Littelmann's LS paths [16] give bases of $H^0(X_\kappa, \mathcal{O}(j))$. Theorem 1.1(a) follows quickly from the latter.[1] Chirivì's semi-toric degenerations [5] and the recent work of Chirivì, Costa Cesari, Fang and Littelmann [6] relate LS paths on chains with prescribed bonds to lecture hall polytopes; their Gorenstein results concern interior LS paths on each toric component. We did not find in these works a statement about Schubert varieties in LG specifically, or about $\chi(X_\kappa, \mathcal{O}(-j))$. Stembridge's enriched P-partitions [27] satisfy a *self*-reciprocity, which is a different statement. For OG, Theorem 1.3 is classical (Remark 4.13). The K-theoretic Pieri rule we use is that of Buch and Ravikumar [4]. To the best of our knowledge, Theorem 1.1(b), Theorem 4.7 as a closed formula, and the half-integer results (Theorem 1.5) have not appeared before.

[1] R. Dehy [8], as reported by Mathieu [20, §4], equips $\{u \le w\}$ with a simplicial structure carrying an integral structure $P_w(n)$, and uses LS paths to interpret dimensions of Demazure modules as Ehrhart polynomials. Mathieu records that it was open whether P_w can always be realised as a lattice polytope. We have not read [8] first-hand. As far as Mathieu's summary shows, it does not contain the lattice Λ or Theorem 1.1(b), but it may anticipate the Ehrhart viewpoint of §3.

1.5. **Outline.** Section 2 fixes notation and lists the cited inputs. Section 3 gives the LS-path proof of Theorem 1.1. Section 4 gives the K-theoretic proof, including the closed form of the Pieri coefficients and the OG case. Section 5 proves Theorem 1.4 in two ways. Section 6 treats half-integers: Theorem 1.5, Corollary 1.6, and the hyperplane-section proof of the vanishing half. Section 7 collects examples, including a complete table for $\mathrm{LG}(3,6)$. Section 8 lists open questions.

2. Preliminaries

2.1. **Shifted diagrams.** Let $\Delta_m = \{(a,b) \in \mathbb{Z}^2 : 0 \le a \le b \le m-1\}$, ordered by the product order. We call a the row and b the column of (a,b), and we draw row a above row $a+1$. Then $x \le y$ means that x is weakly north-west of y. For a strict partition κ with $\kappa_1 \le m$, $D(\kappa) \subseteq \Delta_m$ is the shifted diagram defined in the introduction, and $P_\kappa = (D(\kappa), \le)$. The diagonal box (a,a) is the leftmost box of row a. For $\rho \subseteq \kappa$ (meaning $\rho_i \le \kappa_i$ for all i), row a of $D(\rho)$ is an initial segment of row a of $D(\kappa)$.

Lemma 2.1. *The map $\kappa \mapsto D(\kappa)$ is a bijection from strict partitions with parts at most m to order ideals of Δ_m, and $\rho \subseteq \kappa$ iff $D(\rho) \subseteq D(\kappa)$. Consequently, for $\rho \subseteq \kappa$ the set $\theta = D(\kappa) \setminus D(\rho)$ is an up-set of P_κ, and every up-set of P_κ arises in this way from a unique ρ.*

Proof. $D(\kappa)$ is clearly an order ideal. Conversely, let I be an order ideal. If $(a,b) \in I$ and $b > a$, then $(a,b-1) \in I$, so each row of I is an interval starting on the diagonal. If $(a,b) \in I$ with $a \ge 1$, then $(a-1,b) \in I$, so row $a-1$ reaches at least column b, that is, it is strictly longer than row a. Hence the row lengths form a strict partition. The remaining statements follow, since the complement of an up-set of P_κ is an order ideal of P_κ, hence of Δ_m. □

Lemma 2.2. *$r(\kappa) = 1 + \max\{a+b : (a,b) \in D(\kappa)\} = \max_{1 \le i \le \ell(\kappa)}(\kappa_i + 2i - 2)$, and $r(\varnothing) = 0$.*

Proof. Along a chain $x_1 < \cdots < x_r$ the coordinate sum strictly increases, so $r \le 1 + \max(a+b)$. Conversely, for $(a,b) \in D(\kappa)$ the staircase path $(0,0),(0,1),(1,1),(1,2),\dots,(a,a),(a,a+1),\dots,(a,b)$ consists of $a+b+1$ boxes. Each of them lies below (a,b), so all of them lie in the order ideal $D(\kappa)$, and the path is a chain. The last box of row $i-1$ is $(i-1, i+\kappa_i-2)$, with coordinate sum $\kappa_i + 2i - 3$. □

For example, $r(4,3,2,1) = 7$ and $r(3,2) = 4$. Write $\kappa' = (\kappa_2, \kappa_3, \dots)$. The map $(a,b) \mapsto (a-1, b-1)$ identifies the rows ≥ 1 of P_κ with $P_{\kappa'}$, preserving order and diagonal boxes.

2.2. **Counting functions.** For a finite poset P and $n \ge 0$, let $\Omega(P,n)$ (resp. $\overline{\Omega}(P,n)$) be the number of order-preserving (resp. strictly order-preserving) maps $P \to [n] = \{1,\dots,n\}$. For $D \subseteq P$, let

$$N(P, D; n) = \#\{f \colon P \to [n] \text{ strictly order-preserving} : f(D) \subseteq 2\mathbb{Z}\}.$$

For strict $\nu \subseteq \kappa$ and $j \ge 1$ put

$$N_{\kappa/\nu}(j) = N\big(D(\kappa) \setminus D(\nu),\ \text{diagonal boxes};\ 2j-1\big), \qquad N_\kappa(j) = N_{\kappa/\varnothing}(j),$$

so $N_\varnothing \equiv 1$. For $N \ge 0$ let $F_\kappa(N)$ be the number of order-preserving $g \colon P_\kappa \to \{0,\dots,N\}$ that are even on diagonal boxes.

Lemma 2.3. *Let $\kappa \ne \varnothing$ and $r = r(\kappa)$.*

(a) $N_\kappa(j) > 0$ iff $j \ge \lceil r/2 \rceil + 1$, and $N_\kappa(j) \le N_\kappa(j+1)$ for all $j \ge 1$.
(b) $\overline{\Omega}(P_\kappa, n) > 0$ iff $n \ge r$, and $\overline{\Omega}(P_\kappa, n) \le \overline{\Omega}(P_\kappa, n+1)$.

Proof. (a) The map $f(a,b) = a+b+2$ is strictly order-preserving with values in $[2, r+1]$ (Lemma 2.2), and $f(a,a) = 2a+2$ is even. Conversely, along the staircase chain of Lemma 2.2, $f(0,0) \geq 2$ because $f(0,0)$ is even, and f increases strictly over r boxes, so its last value is at least $r+1$. Hence $N_\kappa(j) > 0$ iff $2j-1 \geq r+1$. Because $2j-1$ is odd, this is $j \geq r/2+1$ for r even and $j \geq (r+3)/2$ for r odd, that is, $j \geq \lceil r/2 \rceil + 1$. Monotonicity holds because $[2j-1] \subseteq [2j+1]$, and the inclusion preserves parity. (b) is the same argument with $f(a,b) = a+b+1$ and no parity constraint. □

2.3. **Reciprocity.** Let $Q \subseteq \mathbb{R}^d$ be a d-dimensional polytope whose vertices lie in a lattice $\Gamma \subseteq \mathbb{R}^d$ of rank d. Then $L_Q(t) = \#(tQ \cap \Gamma)$ is a polynomial in $t \in \mathbb{Z}_{\geq 0}$ of degree d. *Ehrhart–Macdonald reciprocity* [18, 2] says that $L_Q(-t) = (-1)^d \#(\mathrm{int}(tQ) \cap \Gamma)$ for $t \geq 1$. (For $\Gamma \neq \mathbb{Z}^d$, apply a linear isomorphism $\Gamma \cong \mathbb{Z}^d$.) The order polytope of a finite poset P is

$$\mathcal{O}(P) = \{y \in [0,1]^P : x \leq x' \Rightarrow y_x \leq y_{x'}\}.$$

It is full-dimensional, and its vertices are the indicator vectors of the up-sets of P [25]. Its interior is cut out by the strict versions of the facet inequalities, namely $0 < y_x < 1$ and $y_x < y_{x'}$ for $x < x'$. *Stanley's reciprocity* [24], [26, §3.15] says that $\Omega(P, -n) = (-1)^{|P|}\overline{\Omega}(P, n)$ for $n \geq 0$. **[CHECK: theorem number in EC1, 2nd ed., not verified; cited by section.]**

2.4. **Schubert varieties.** *Type C.* Let $G = \mathrm{Sp}_{2m}$ with ε-coordinates, simple roots $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$ $(i < m)$ and $\alpha_m = 2\varepsilon_m$, and $\omega = \omega_m = \varepsilon_1 + \cdots + \varepsilon_m$. Coroots are $(\varepsilon_i \pm \varepsilon_k)^\vee = \varepsilon_i \pm \varepsilon_k$ and $(2\varepsilon_i)^\vee = \varepsilon_i$. Let $P \supseteq B$ be the maximal parabolic subgroup with $W_P = \mathrm{Stab}_W(\omega) \cong S_m$. Then $G/P = \mathrm{LG}(m, 2m)$, and $L_\omega = \mathcal{O}(1)$ is the ample generator of the Picard group. The orbit $W\omega$ is $\{\pm 1\}^m \cong W/W_P$. For a sign vector μ, let $w_\mu \in W^P$ be the minimal coset representative with $w_\mu \omega = \mu$, and let $X_\mu = \overline{Bw_\mu P/P}$, of dimension $\ell(w_\mu)$. Put $\kappa(\mu) = \{m - i + 1 : \mu_i = -1\}$, a strict partition with parts at most m. By Lemma 3.1 below, $\mu \mapsto \kappa(\mu)$ is a bijection and $\dim X_\mu = |\kappa(\mu)|$. We write $X_\kappa = X_\mu$ and $w_\kappa = w_\mu$ when $\kappa = \kappa(\mu)$. Thus $w_\kappa \omega = v_\kappa$, the sign vector with -1 exactly in positions $m + 1 - k$, $k \in \kappa$.

Type B. Let $G = \mathrm{Spin}_{2m+1}$ with $\alpha_m = \varepsilon_m$ (short), $\alpha_m^\vee = 2\varepsilon_m$, and $\omega_m = \frac{1}{2}(\varepsilon_1 + \cdots + \varepsilon_m)$. Then $G/P_m = \mathrm{OG}(m, 2m+1)$ and L_{ω_m} is the spinor line bundle, the ample generator. Schubert varieties X_κ^{OG} are indexed in the same way.

For either type, $H_\kappa(j) = \chi(X_\kappa, L^j)$ is a polynomial in j of degree $|\kappa|$ [13], and $H_\varnothing \equiv 1$ because $X_\varnothing$ is a point.

Remark 2.4 (Two indexings). Buch and Ravikumar [4] index Schubert varieties by codimension: X^λ has codimension $|\lambda|$, and we put $\lambda^\vee = [m] \setminus \lambda$ (as sets of parts). In §4 we use that X_κ is a G-translate of $X^{\kappa^\vee}$, so the two have the same Hilbert polynomial and the same class in $K(X)$. **[CHECK: this dictionary is used but not derived from BR's definitions in the source notes. It is confirmed computationally: the K-class $[\mathcal{O}_{X_\kappa}] \cdot \mathcal{O}^1$ computed from Demazure characters in the convention above agrees with BR's formula in all 46 cases checked ($m \leq 5$), and the Demazure Hilbert polynomials agree with the strict counts of Theorem 1.1(b) at $m \leq 5$. Note that the poset of shifted diagrams alone does not fix the dictionary: for $m = 3$ it has an automorphism exchanging (3) and $(2,1)$, whose Hilbert polynomials differ.]**

2.5. **Demazure operators and cited inputs.** For a simple root α_i, the Demazure operator on $\mathbb{Z}[\text{weights}]$ is $D_i f = (f - e^{-\alpha_i} s_i f)/(1 - e^{-\alpha_i})$. For $n = \langle \lambda, \alpha_i^\vee \rangle$,

$$D_i e^\lambda = \begin{cases} e^\lambda + e^{\lambda - \alpha_i} + \cdots + e^{s_i \lambda}, & n \geq 0, \\ 0, & n = -1, \\ -(e^{\lambda + \alpha_i} + \cdots + e^{s_i \lambda - \alpha_i}), & n \leq -2. \end{cases}$$

For a reduced word $w = s_{i_1} \cdots s_{i_k}$, put $D_w = D_{i_1} \circ \cdots \circ D_{i_k}$; it does not depend on the reduced word. Littelmann's operator $\Lambda_\alpha(e^\mu) = (e^{\mu+\rho} - e^{s_\alpha(\mu+\rho)})/(1-e^{-\alpha}) \cdot e^{-\rho}$ equals D_α, because $s_\alpha(\mu+\rho) = s_\alpha\mu + \rho - \alpha$ for simple α.

The following inputs are standard. We cite them and do not reprove them.

(G0) (Chevalley.) $\operatorname{Pic} X = \mathbb{Z}L$, and the unique Schubert divisor $X^{(1)}$ is the reduced zero scheme of a section of L. Hence $\mathcal{O}^1 := [\mathcal{O}_{X^{(1)}}] = 1 - [L^{-1}]$ in $K(X)$.

(G1) (Demazure character formula [9, 12, 1, 22, 14, 19]; see [3, Ch. 3].) For $w \in W^P$ and dominant λ with $W_P \subseteq \operatorname{Stab}_W(\lambda)$, $H^i(X_w, L_\lambda) = 0$ for $i > 0$, and $\operatorname{char} H^0(X_w, L_\lambda)^* = D_w(e^\lambda)$.

(LS) (Littelmann [16, §5.2]; see [17, §8, p. 305].) For dominant λ and $w \in W/W_\lambda$, $D_w(e^\lambda) = \sum_\pi e^{\pi(1)}$, where the sum runs over LS paths π of shape λ with initial direction $\le w$. This holds for every symmetrizable Kac–Moody algebra, in particular in type C_m. **[CHECK: the §5.2 locator in Invent. Math. 116 is secondary (quoted from Chirivì–Fang–Littelmann, arXiv:2207.08904); the statement itself was read in the ICM 1994 survey, p. 305.]**

(BR) (Buch–Ravikumar [4].) The K-theoretic Pieri rule for $\mathrm{LG}(m, 2m)$ and $\mathrm{OG}(m, 2m+1)$; the precise statements used are quoted in §4.2.

Combining (G1) with (LS) gives $h^0(X_\kappa, \mathcal{O}(j)) = \#\{\text{LS paths of shape } j\omega \text{ with initial direction} \le w_\kappa\}$. Only this count is used in §3.

The next lemma lets us evaluate Demazure expressions at non-integral weights. Work with analytic germs at $h = 0$ on $\mathfrak{h} = \mathbb{C}^m$, with $e^\mu(h) = e^{\langle \mu, h\rangle}$ for $\mu \in \mathbb{C}^m$. Each D_i maps germs to germs, because the numerator vanishes on $\alpha_i(h) = 0$ and $1 - e^{-\alpha_i}$ is α_i times a unit.

Lemma 2.5 (Polynomial extension)**.** *For $w \in W$, $P_w(\lambda) := [D_w e^\lambda](0)$ is a polynomial in $\lambda \in \mathbb{C}^m$ of degree at most $\ell(w)$. Given (G1), $H_\kappa(j) = P_{w_\kappa}(j\omega)$ for every $j \in \mathbb{C}$.*

Proof. Let $\mathfrak{m}$ be the maximal ideal of germs at 0. For $f \in \mathfrak{m}^k$ with $k \ge 1$,

$$D_i f = \frac{f - s_i f}{1 - e^{-\alpha_i}} + s_i f,$$

and $(f - s_i f)/\alpha_i \in \mathfrak{m}^{k-1}$. Hence $D_i(\mathfrak{m}^k) \subseteq \mathfrak{m}^{k-1}$ and $D_w(\mathfrak{m}^{\ell(w)+1}) \subseteq \mathfrak{m}$. Expanding $e^\lambda = \sum_{k \le \ell(w)} \langle \lambda, h\rangle^k / k! + R$ with $R \in \mathfrak{m}^{\ell(w)+1}$ shows that $P_w(\lambda)$ is a polynomial in λ. For integral λ, $D_w e^\lambda$ is a Laurent polynomial, and its value at $h = 0$ is its coefficient sum. By (G1), $H_\kappa(j) = P_{w_\kappa}(j\omega)$ for integers $j \ge 0$, and both sides are polynomials in j. □

Lemma 2.6 (The anchor)**.** *If $\kappa \ne \varnothing$, then $H_\kappa(-1) = 0$, both on* LG *and on* OG.

Proof. Let $w = w_\kappa$. Since w is minimal in wW_P, $ws_i > w$ for every $i \ne m$. Since $w \ne e$, it has a right descent, so $ws_m < w$, and we may write $w = w's_m$ with lengths adding. Then $D_w = D_{w'} D_m$, and D_m acts first. In both types $\langle -\omega_m, \alpha_m^\vee \rangle = -1$, so $D_m e^{-\omega_m} = 0$. By Lemma 2.5, $H_\kappa(-1) = P_w(-\omega_m) = 0$. □

3. The LS-path proof of strict reciprocity

This section proves Theorem 1.1 using only (G1) and (LS). It uses neither the Buch–Ravikumar Pieri rule nor (G0).

3.1. **LS paths.** We recall Littelmann's definition [16, 17]. Let λ be dominant, and give $W^\lambda = W/W_\lambda$ the Bruhat order. For $\sigma > \tau$ in W^λ and a rational $a \in (0,1)$, an *a-chain* for (σ, τ) is a sequence of Bruhat covers $\sigma = \kappa_0 > \kappa_1 > \cdots > \kappa_r = \tau$ with $\kappa_{k-1} = s_{\beta_k}\kappa_k$ ($\beta_k > 0$) and $a\langle \kappa_k(\lambda), \beta_k^\vee \rangle \in \mathbb{Z}$ for all k. An *LS path of shape λ* is a pair

$$\pi = (\tau_1 > \tau_2 > \cdots > \tau_s \ ; \ 0 = a_0 < a_1 < \cdots < a_s = 1)$$

such that an a_i-chain for (τ_i, τ_{i+1}) exists for $1 \le i \le s-1$. Its *initial direction* is τ_1, and $\pi(1) = \sum_i (a_i - a_{i-1})\tau_i(\lambda)$. We call $|\langle \kappa_k(\lambda), \beta_k^\vee \rangle|$ the *bond* of the cover $\kappa_{k-1} > \kappa_k$. The absolute value makes it independent of sign conventions, because $\langle s_\beta \mu, \beta^\vee \rangle = -\langle \mu, \beta^\vee \rangle$.

3.2. **Covers and bonds.**

Lemma 3.1. *In type C_m with $\lambda = \omega_m$:*

(a) $\dim X_\mu = \#\{\alpha > 0 : \langle \mu, \alpha^\vee \rangle < 0\} = \sum_{i:\mu_i=-1}(m-i+1) = |\kappa(\mu)|$;

(b) the Bruhat covers $\mu < s_\beta \mu$ in W^λ are exactly the moves for which $D(\kappa(s_\beta\mu))$ is $D(\kappa(\mu))$ plus one box;

(c) the bond of such a cover is 1 if the added box is diagonal and 2 otherwise.

In particular, the Bruhat order on W^λ is inclusion of shifted diagrams.

Proof. (a) The positive roots are $\varepsilon_i \pm \varepsilon_k$ $(i < k)$ and $2\varepsilon_i$. The pairing $\langle \mu, \varepsilon_i - \varepsilon_k \rangle$ is negative iff $\mu_i = -1$ and $\mu_k = +1$; $\langle \mu, \varepsilon_i + \varepsilon_k \rangle < 0$ iff $\mu_i = \mu_k = -1$; and $\langle \mu, \varepsilon_i \rangle < 0$ iff $\mu_i = -1$. So each i with $\mu_i = -1$ contributes $1 + (m-i)$.

(b, c) Let $i < k$, $p = m - i + 1$ and $q = m - k + 1$, so $p > q$. The reflection $s_{\varepsilon_i - \varepsilon_k}$ swaps μ_i and μ_k. Upward $(\mu_i = +1, \mu_k = -1)$ it replaces the part q by p and raises the dimension by $p - q$. It is a cover iff $p = q + 1$, that is, iff it adds one box at the end of the row of length q. That box is off the diagonal because $q \ge 1$, and the bond is $|1 - (-1)| = 2$. The reflection $s_{\varepsilon_i + \varepsilon_k}$ flips μ_i, μ_k when they are equal. Upward it adds the two parts p and q, raising the dimension by $p + q \ge 3$, so it is never a cover. The reflection $s_{2\varepsilon_i}$ flips μ_i. Upward it adds the part p, so it is a cover iff $p = 1$ (that is, $i = m$): a new row of length one, whose box is diagonal. Its bond is $|\langle \mu, \varepsilon_m \rangle| = 1$. Every cover has the form $\mu < s_\beta \mu$ with the dimension rising by one, so the list is complete. The Bruhat order is the transitive closure of its covers, and adding boxes one at a time (along any linear extension of $D(\kappa_2) \setminus D(\kappa_1)$) generates inclusion of shifted diagrams. □

3.3. **LS paths of shape $j\omega_m$.**

Lemma 3.2. *Let $j \ge 1$. A pair $(\tau_1 \supsetneq \cdots \supsetneq \tau_s\,;\, 0 < a_1 < \cdots < a_{s-1} < 1)$ of shifted diagrams and rationals is an LS path of shape $j\omega_m$ iff, for every $i \le s-1$, $2ja_i \in \mathbb{Z}$, and if $2ja_i$ is odd, then $D(\tau_i) \setminus D(\tau_{i+1})$ contains no diagonal box.*

Proof. Since $W_{j\omega_m} = W_{\omega_m}$, the poset is unchanged, and all bonds are multiplied by j. So the a-chain condition reads $a\, j\, b \in \mathbb{Z}$ for the bond $b \in \{1, 2\}$ of each cover in the chain. If an a-chain exists, it has at least one cover, so $2ja \in \mathbb{Z}$. If $ja \in \mathbb{Z}$, then every saturated chain from τ_{i+1} up to τ_i is an a-chain, and one exists by Lemma 3.1. If $2ja$ is odd, a cover qualifies iff its bond is 2, iff it adds an off-diagonal box. If $D(\tau_i) \setminus D(\tau_{i+1})$ has no diagonal box, every saturated chain adds only off-diagonal boxes. If it has one, every saturated chain adds it at some step. □

Proposition 3.3 (The LS-parity model)**.** *For strict κ with parts at most m and $j \ge 0$, the LS paths of shape $j\omega_m$ with initial direction $\le \kappa$ are in bijection with the order-preserving maps $G\colon P_\kappa \to \{0, \dots, 2j\}$ that are even on diagonal boxes. Hence, given (G1) and (LS), $H_\kappa(j) = h^0(X_\kappa, \mathcal{O}(j)) = F_\kappa(2j)$ for all $j \ge 0$.*

Proof. For $j = 0$ both sides equal 1: the constant path, and $G \equiv 0$. Let $j \ge 1$ and let $\pi = (\tau_1 > \cdots > \tau_s; a)$ be an LS path with $\tau_1 \le \kappa$, so $D(\tau_1) \subseteq D(\kappa)$ by Lemma 3.1. For $x \in D(\kappa)$ let $i(x) = \max\{i : x \in D(\tau_i)\}$ and $t(x) = a_{i(x)}$, with $t(x) = 0$ if $x \notin D(\tau_1)$. Set $g(x) = 2j\, t(x) \in \{0, \dots, 2j\}$; this is an integer by Lemma 3.2.

- *g is order-reversing.* The $D(\tau_i)$ are order ideals. If $y \le x$ and $x \in D(\tau_i)$, then $y \in D(\tau_i)$, so $i(y) \ge i(x)$ and $g(y) \ge g(x)$.

- *g is even on the diagonal.* A diagonal box x with $1 \le i(x) \le s-1$ lies in $D(\tau_{i(x)}) \setminus D(\tau_{i(x)+1})$, so $g(x) = 2ja_{i(x)}$ is even by Lemma 3.2. The values 0 and $2j$ are even anyway.

Conversely, let $g\colon P_\kappa \to \{0,\dots,2j\}$ be order-reversing and even on the diagonal. Let $v_1 < \cdots < v_r$ be its distinct values in the open interval $(0,2j)$. Put $a_i = v_i/2j$ and $\tau_i = \{x : g(x) \ge v_i\}$ for $i \le r$, and $\tau_{r+1} = \{x : g(x) = 2j\}$. Each τ_i is an order ideal of P_κ, hence a shifted diagram by Lemma 2.1. They decrease strictly, because $\tau_i \setminus \tau_{i+1} = g^{-1}(v_i) \neq \varnothing$. A diagonal box in $g^{-1}(v_i)$ forces v_i to be even, so Lemma 3.2 applies and we get an LS path with initial direction $\tau_1 \le \kappa$. (If $\tau_{r+1} = \varnothing$, the last direction is the identity coset, which is allowed; if $r = 0$ the path has the single direction $g^{-1}(2j)$.) The two constructions are inverse, because $x \in \tau_i \setminus \tau_{i+1}$ iff $g(x) = v_i$ iff $t(x) = a_i$. Finally, $G = 2j - g$ is order-preserving and even on the diagonal, since $2j$ is even. The count $h^0 = \#\{\text{paths}\}$ is (G1) combined with (LS), and $h^0 = \chi$ by (G1). □

3.4. **Proof of Theorem 1.1.** Let $\Lambda \subseteq \mathbb{R}^{P_\kappa}$ be the lattice with $\mathbb{Z}$ on diagonal coordinates and $\frac{1}{2}\mathbb{Z}$ on the others. The order polytope $\mathcal{O} = \mathcal{O}(P_\kappa)$ has its vertices in $\mathbb{Z}^{P_\kappa} \subseteq \Lambda$. Rescaling the off-diagonal coordinates by 2 maps Λ onto $\mathbb{Z}^{P_\kappa}$ and $\mathcal{O}$ onto a lattice polytope, so $L(j) = \#(j\mathcal{O} \cap \Lambda)$ is a polynomial in j of degree $|\kappa|$. The map $G \mapsto y = G/2$ is a bijection from the maps counted by $F_\kappa(2j)$ onto $j\mathcal{O} \cap \Lambda$: the diagonal coordinate y_x is an integer iff $G(x)$ is even. Hence $L(j) = F_\kappa(2j)$ for $j \ge 0$. (As a function of N, $F_\kappa(N)$ is only a quasi-polynomial of period 2; we use it only at even N.)

By Proposition 3.3, the polynomials H_κ and L agree at every $j \ge 0$, so $H_\kappa = L$. This proves (a) and (c). The interior of $j\mathcal{O}$ is $\{0 < y_x < j,\ y_x < y_{x'} \text{ for } x < x'\}$. Under $y = f/2$, its points in Λ are exactly the strictly order-preserving $f\colon P_\kappa \to [2j-1]$ with even diagonal values. Ehrhart–Macdonald reciprocity for $(\mathcal{O}, \Lambda)$ gives

$$H_\kappa(-j) = L(-j) = (-1)^{|\kappa|}\#(\mathrm{int}(j\mathcal{O}) \cap \Lambda) = (-1)^{|\kappa|}N_\kappa(j) \qquad (j \ge 1),$$

which is (b). □

Example 3.4. For $\kappa = (1)$ the LS paths with $\tau_1 \le (1)$ are: the constant paths $\varnothing$ and (1), and the paths $(1) > \varnothing$ with time $p/2j$. The box is diagonal, so p must be even, which leaves $j-1$ choices. In total there are $j+1 = H_{(1)}(j)$ paths. For $\kappa = (2)$ and $j = 1$ the paths are $\varnothing$, (1), (2) and $(2) > (1)$ at time $\frac{1}{2}$. The last one is allowed because the box $(0,1)$ is off-diagonal. The paths $(1) > \varnothing$ and $(2) > \varnothing$ at time $\frac{1}{2}$ are not allowed. This gives $4 = H_{(2)}(1) = (1+1)^2$.

Remark 3.5 (The orthogonal case). On $\mathrm{OG}(m, 2m+1)$ the weight ω_m is minuscule, so every bond is 1. The analogue of Lemma 3.2 then forces $ja_i \in \mathbb{Z}$, and the same bijection gives $H^{\mathrm{OG}}_\kappa(j) = \Omega(P_\kappa, j+1)$. We have checked by computer that all type-B covers in W/W_{ω_m} have bond 1, and that the LS count equals $\Omega(P_\kappa, j+1)$ and the type-B Demazure character count in 72 cases ($m \le 4$, $j \le 3$). We have not written out the type-B analogue of Lemma 3.1, so this route is recorded as a remark only. The statement itself is classical (Remark 4.13).

Remark 3.6 (Computer checks). LS paths of shape $j\omega_m$ were enumerated directly from the root data, with covers, bonds and a-chains computed from roots rather than from the box dictionary. Their full weight multisets $\sum e^{\pi(1)}$ were compared with Demazure characters, and their number with $F_\kappa(2j)$: 164 cases ($m = 2,3$ with $j \le 3$; $m = 4$ with $j \le 4$; $m = 5$ with $j \le 2$), with no failures. Two negative controls break the bond rule: setting every bond to 2, and swapping the bonds. Both fail against the Demazure characters in most cases (for example 30/32 and 28/32 failures at $m = 4$).

4. The K-theoretic proof

This section gives a second proof of Theorem 1.1(b), using (G0), (G1) and (BR) but not (LS). Its main ingredient, Theorem 4.7, is a closed formula for the K-theoretic Pieri coefficients of the Schubert divisor. That formula is reused in §§5–6.

4.1. The hyperplane recursion.

Lemma 4.1. *Let $X = \mathrm{LG}(m, 2m)$ or $\mathrm{OG}(m, 2m+1)$ and write*

$$[\mathcal{O}_{X_\kappa}] \cdot (1 - [L^{-1}]) = \sum_\rho c_\rho \, [\mathcal{O}_{X_\rho}] \qquad \text{in } K(X).$$

Then $c_\rho \neq 0$ implies $\rho \subsetneq \kappa$, and for every $j \in \mathbb{C}$,

$$H_\kappa(j) - H_\kappa(j-1) = \sum_{\rho \subsetneq \kappa} c_\rho \, H_\rho(j). \tag{1}$$

By (G0), c_ρ is the coefficient of $[\mathcal{O}_{X_\rho}]$ in $[\mathcal{O}_{X_\kappa}] \cdot \mathcal{O}^1$.

Proof. Since L is very ample and X_κ is integral, there is a section s of L that does not vanish identically on X_κ. Let $Z \subseteq X_\kappa$ be its zero scheme. Then $[\mathcal{O}_{X_\kappa}] - [\mathcal{O}_{X_\kappa} \otimes L^{-1}] = [\mathcal{O}_Z]$. The variety X_κ is the disjoint union of the Schubert cells C_ρ, $\rho \subseteq \kappa$, and C_κ is open in X_κ. By the localization sequence $K_0(X_\kappa \setminus C_\kappa) \to K_0(X_\kappa) \to K_0(C_\kappa) \to 0$ and induction, the image of $K_0(X_\kappa)$ in $K(X)$ is spanned by the $[\mathcal{O}_{X_\rho}]$, $\rho \subseteq \kappa$. Restriction to $C_\kappa \cong \mathbb{A}^{|\kappa|}$ followed by the rank gives a map $K_0(X_\kappa) \to K_0(C_\kappa) \cong \mathbb{Z}$. It sends $[\mathcal{O}_{X_\kappa}] \mapsto 1$ and $[\mathcal{O}_{X_\rho}] \mapsto 0$ for $\rho \subsetneq \kappa$. It also sends $[\mathcal{O}_Z] \mapsto 0$, because $Z \cap C_\kappa$ is a proper closed subset of the irreducible C_κ. Hence the coefficient of $[\mathcal{O}_{X_\kappa}]$ vanishes, and $c_\rho \neq 0$ forces $\rho \subsetneq \kappa$. Multiplying by $[L^j]$ and applying χ gives (1) for integers j. Both sides are polynomials in j, so it holds for all j. □

4.2. The Buch–Ravikumar Pieri rule.

We quote the statements we use from [4]. Their numbering below follows the arXiv source of [4] (§4 is OG, §5 is LG). **[CHECK: theorem numbering taken from the arXiv v1 LaTeX source of 1005.2605 (theorem counter reconstructed by hand); not compared with the published version in J. reine angew. Math.]**

In [4] Schubert varieties are indexed by codimension: $\mathcal{O}^\lambda = [\mathcal{O}_{X^\lambda}]$ and $\mathcal{O}^\lambda \cdot \mathcal{O}^p = \sum_\nu c^\nu_{\lambda,p} \mathcal{O}^\nu$. A shifted skew diagram θ is a *rim* if no box of θ lies strictly south and strictly east of another box of θ. If θ is a nonempty rim, its *upper-right box* is the rightmost box B of its top row. Its *north-east arm* ψ is the largest row or column that can be obtained by intersecting θ with a square whose upper-right box is B. Put $\hat\theta = \theta \setminus \psi$; then $\hat\theta = \varnothing$ iff θ is a row or a column. Two boxes are *connected* if they share a side.

- [4, Cor. 5.4] For strict $\lambda \subset \nu$ with $\nu_1 \leq m$ and $0 \leq p \leq m$, $c^\nu_{\lambda,p} = \mathcal{C}(\nu/\lambda, p)$. Here $\mathcal{C}$ is an explicit alternating sum [4, eq. (11)] over subsets of the south-east corners.
- [4, Thm. 5.5] If θ is not a rim, then $\mathcal{C}(\theta, p) = 0$. If $p \leq 0$, then $\mathcal{C}(\theta, p) = \delta_{\theta,\varnothing}$. If $\theta \neq \varnothing$ is a rim, $p > 0$ and $a = |\theta| - |\hat\theta|$, then:
  - (i) if $\hat\theta = \varnothing$ and θ meets the diagonal, then $\mathcal{C}(\theta, p) = \delta_{p,|\theta|} - \delta_{p,|\theta|-1}$ if θ is a column, and $\mathcal{C}(\theta, p) = \delta_{p,|\theta|}$ otherwise;
  - (ii) if $\hat\theta = \varnothing$ and θ is disjoint from the diagonal, then $\mathcal{C}(\theta, p) = 2\delta_{p,|\theta|} - \delta_{p,|\theta|-1}$;
  - (iii) if $\hat\theta \neq \varnothing$ and ψ is connected to $\hat\theta$, then $\mathcal{C}(\theta, p) = \mathcal{C}(\hat\theta, p-a) - \mathcal{C}(\hat\theta, p-a+1)$;
  - (iv) if $\hat\theta \neq \varnothing$ and ψ is not connected to $\hat\theta$, then $\mathcal{C}(\theta, p) = 2\mathcal{C}(\hat\theta, p-1) - 2\mathcal{C}(\hat\theta, p)$ for $a = 1$, and $\mathcal{C}(\theta, p) = 2\mathcal{C}(\hat\theta, p-a) - 3\mathcal{C}(\hat\theta, p-a+1) + \mathcal{C}(\hat\theta, p-a+2)$ for $a > 1$.

- [4, Cor. 4.4, Lemma 4.5] (OG.) For $\lambda \subset \nu$, $c^{\nu}_{\lambda,p} = \mathcal{B}(\nu/\lambda, p)$. If θ is not a rim, then $\mathcal{B}(\theta, p) = 0$; if θ contains a row or column with a boxes, then $\mathcal{B}(\theta, p) = 0$ for $p < a$; and if θ is a rook strip (at most one box in each row and each column), then $\mathcal{B}(\theta, 1) = (-1)^{|\theta|-1}$.

Buch and Ravikumar prove all of these. They also state a tableau formula for $c^{\nu}_{\lambda,p}$ on LG in terms of KLG-tableaux [4, Cor. 5.6], whose proof they leave to the reader. We do not use it: Proposition 4.6 below proves the case $p = 1$ from Thm. 5.5.

Lemma 4.2 (Reflection)**.** *Let $R(a, b) = (m - 1 - b, m - 1 - a)$, an involution of Δ_m. Then $D(\kappa^\vee) = R(\Delta_m \setminus D(\kappa))$. Consequently, for $\rho \subseteq \kappa$,*

$$D(\kappa) \setminus D(\rho) = R(D(\rho^\vee) \setminus D(\kappa^\vee)),$$

and $\rho \subsetneq \kappa$ iff $\kappa^\vee \subsetneq \rho^\vee$. The map R reverses the product order, preserves the diagonal, and exchanges rows with columns.

Proof. Take $(a, b) \in \Delta_m$ and put $d = b - a + 1$. Then $(a, b) \in D(\kappa)$ iff $\kappa_{a+1} \geq d$, iff $\#(\kappa \cap [d, m]) \geq a + 1$. Write $R(a, b) = (\bar{a}, \bar{b})$ with $\bar{a} = m - 1 - b$; then $\bar{b} - \bar{a} + 1 = d$. By the same criterion, $R(a, b) \in D(\kappa^\vee)$ iff $\#(\kappa^\vee \cap [d, m]) \geq m - b$. Since $\#(\kappa \cap [d, m]) + \#(\kappa^\vee \cap [d, m]) = m - d + 1 = m - b + a$, this says $\#(\kappa \cap [d, m]) \leq a$, that is, $(a, b) \notin D(\kappa)$. □

So, with X_κ a translate of $X^{\kappa^\vee}$ (Remark 2.4), the coefficient in Lemma 4.1 is $c_\rho = c^{\rho^\vee}_{\kappa^\vee,1} = \mathcal{C}(\theta_{BR}, 1)$, where $\theta_{BR} = D(\rho^\vee) \setminus D(\kappa^\vee) = R(\theta)$ and $\theta = D(\kappa) \setminus D(\rho)$.

4.3. **The Pieri coefficients count two-letter strict maps.** For a finite poset P and $D \subseteq P$, let $V(P, D)$ be the number of strictly order-preserving $f\colon P \to \{e < o\}$ with $f(D) \subseteq \{e\}$. For an up-set θ of P_κ, write $V(\theta) = V(\theta, \text{diagonal boxes of } \theta)$.

Lemma 4.3. *$V(P, D) = 0$ if P contains a chain $x < y < z$, or if some $d \in D$ is not minimal in P. Otherwise $V(P, D) = 2^i$, where i is the number of elements of $P \setminus D$ that are comparable to no other element.*

Proof. A 3-chain needs three distinct values. If $x < d \in D$, then $f(x) < f(d) = e$ is impossible. Otherwise P has height at most 2. A non-isolated x is then either non-maximal, which forces $f(x) = e$, or non-minimal, which forces $f(x) = o$, but not both (that would give a 3-chain). These forced values satisfy every relation. A non-isolated $d \in D$ is minimal, so it is forced to e, consistently with D. An isolated element is free (two choices) unless it lies in D. □

In P_κ, a diagonal box (c, c) of an up-set θ is non-minimal in θ iff $(c - 1, c) \in \theta$. Indeed, if some $(a, b) < (c, c)$ lies in θ, then $a \leq c - 1$ and $b \leq c$, so $(a, b) \leq (c - 1, c) < (c, c)$, and $(c - 1, c) \in \theta$ because θ is an up-set.

For a shifted skew shape φ in Buch–Ravikumar's convention, let $K(\varphi)$ be the number of labellings $\varphi \to \{1' < 1\}$ that strictly increase along every row (left to right) and every column (top to bottom) and put no $1'$ on a diagonal box, with $K(\varphi) = 0$ if φ is not a rim.

Lemma 4.4. *Let $\varphi = \nu/\lambda$ be a shifted skew shape.*

(a) If φ is not a rim, then φ contains a 3-chain in the product order.

(b) If φ is a rim, then two boxes of φ are comparable iff they share a row or a column.

Proof. (a) Let $x = (r, c)$ and $y = (r', c')$ be boxes of φ with $r < r'$ and $c < c'$. The right ends of the rows of a shifted strict diagram weakly decrease, so $(r, c') \in \nu$. Rows of λ are initial segments and $(r, c) \notin \lambda$, so $(r, c') \notin \lambda$. Hence $x < (r, c') < y$ in φ. (b) Comparable means weakly north-west, and in a rim no pair is strictly north-west, so comparable boxes share a row or a column. □

Lemma 4.5. *For every $\rho \subsetneq \kappa$, $V(\theta) = K(R(\theta))$.*

Proof. If $R(\theta)$ is not a rim, it has a 3-chain by Lemma 4.4(a). Since R reverses the order, θ has one too, and $V(\theta) = 0 = K(R(\theta))$. If $R(\theta)$ is a rim, then by Lemma 4.4(b) "strict along rows and columns" means "strictly order-preserving" on $R(\theta)$. Composing with R (which reverses order and preserves the diagonal) and renaming $1 \mapsto e$, $1' \mapsto o$ gives a bijection. It sends strictly order-preserving maps $R(\theta) \to \{1' < 1\}$ with diagonal boxes labelled 1 to strictly order-preserving maps $\theta \to \{e < o\}$ with diagonal boxes sent to e. □

Proposition 4.6. *For every nonempty shifted skew shape φ, $\mathcal{C}(\varphi, 1) = (-1)^{|\varphi|-1}K(\varphi)$.*

This is the case $p = 1$ of [4, Cor. 5.6]. We prove it from [4, Thm. 5.5] by induction on $|\varphi|$. At $p = 1$, and using $\mathcal{C}(\hat{\varphi}, q) = 0$ for $q \leq 0$ when $\hat{\varphi} \neq \varnothing$, Thm. 5.5 reads as follows:

(i) diagonal column: 1 if $|\varphi| = 1$, -1 if $|\varphi| = 2$, 0 otherwise; diagonal row: 1 if $|\varphi| = 1$, 0 otherwise;
(ii) off-diagonal row or column: 2 if $|\varphi| = 1$, -1 if $|\varphi| = 2$, 0 otherwise;
(iii) connected arm: $-\mathcal{C}(\hat{\varphi}, 1)$ if $a = 1$, and 0 if $a \geq 2$;
(iv) disconnected arm: $-2\mathcal{C}(\hat{\varphi}, 1)$ if $a = 1$, $+\mathcal{C}(\hat{\varphi}, 1)$ if $a = 2$, and 0 if $a \geq 3$.

Proof. If φ is not a rim, both sides vanish. Let φ be a nonempty rim, and write $T(\varphi) = (-1)^{|\varphi|-1}K(\varphi)$. We use the following facts.

(g1) Rows and columns of a shifted skew shape are intervals.
(g2) A diagonal box (c, c) is the leftmost box of its row and the lowest box of its column.
(g3) Let r be the top row of φ and $B = (r, c)$ its rightmost box. Then c is the largest column of φ: a box (r', c') with $c' > c$ would have $r' > r$, and B would be strictly north-west of it. So B's only possible neighbours in φ are $L = (r, c-1)$ and $D = (r+1, c)$.
(g4) L and D are not both in φ, because D is strictly south-east of L.
(g5) In a rim, comparable boxes share a row or a column (Lemma 4.4(b)), so K sees only row and column constraints.

Cases (i), (ii): $\varphi = \psi$ is a row or a column. With two labels, $K = 0$ for length ≥ 3. A diagonal column has its diagonal box at the bottom (g2). Length 1 gives the label 1, so $K = 1 = T$. Length 2 gives $1'$ over 1, so $K = 1$ and $T = -1$. A diagonal row has its diagonal box leftmost. Length 1 gives $K = 1 = T$. In length 2 the leftmost box would need $1'$, which is forbidden, so $K = 0$. Off the diagonal, length 1 gives $K = 2 = T$ and length 2 gives $K = 1$ (labels $1', 1$), so $T = -1$. All of these agree with (i) and (ii).

Cases (iii) and (iv) with $a \geq 3$. Here ψ is a row or column of at least three boxes, so $K(\varphi) = 0 = \mathcal{C}(\varphi, 1)$.

Case (iii) with $a = 1$. Here $\psi = \{B\}$ and B is adjacent to $\hat{\varphi}$, so exactly one of L, D lies in φ (g3, g4). Let $S = (r+1, c-1)$. If $S \notin \varphi$, then the 2×2 square with upper-right box B meets φ in a row or column of two boxes, so $a \geq 2$. Hence $S \in \hat{\varphi}$.

- If $L \in \varphi$: L is directly above S, so every valid labelling of $\hat{\varphi}$ has $L = 1'$. Then B, to the right of L, is forced to be 1, which is allowed. B is alone in its column ($D \notin \varphi$; g1). The boxes of B's row other than B lie in $\hat{\varphi}$, and their constraints are already part of $\hat{\varphi}$. (A box to the left of L would need a label smaller than $1'$; then $K(\hat{\varphi}) = 0 = K(\varphi)$.) So $K(\varphi) = K(\hat{\varphi})$.
- If $D \in \varphi$: S is directly left of D, so $D = 1$ in every labelling of $\hat{\varphi}$. Then B, above D, is forced to be $1'$. This is allowed, because B is off the diagonal: $D = (r+1, c)$ lies in the shifted plane, so $c \geq r + 1$. B is alone in its row ($L \notin \varphi$). Again $K(\varphi) = K(\hat{\varphi})$.

Hence $T(\varphi) = -T(\hat{\varphi}) = -\mathcal{C}(\hat{\varphi}, 1)$ by induction.

Case (iii) with $a = 2$. Here ψ is $\{L, B\}$ (a row) or $\{B, D\}$ (a column), and the 2×2 square at B meets φ in exactly ψ. In the row case, $S, D \notin \varphi$. The neighbours of L are $(r, c-2)$, S, and

the box above L, which is absent because r is the top row. So the connection to $\hat{\varphi}$ goes through $(r, c-2) \in \varphi$. Then row r has three boxes, and $K(\varphi) = 0$. In the column case, $S, L \notin \varphi$. The neighbours of D are $(r+2, c)$, S and $(r+1, c+1)$; the last is absent because column c is maximal. So $(r+2, c) \in \varphi$, column c has three boxes, and $K(\varphi) = 0$. Both agree with $\mathcal{C}(\varphi, 1) = 0$.

Case (iv) with $a \in \{1, 2\}$. Here ψ is not connected to $\hat{\varphi}$.

- No box of ψ shares a row or column with a box of $\hat{\varphi}$. Otherwise, walking along the common row or column interval (g1) from the ψ-box to the $\hat{\varphi}$-box would cross an adjacency between ψ and $\hat{\varphi}$.
- ψ contains no diagonal box. If ψ is a row in row r containing (r, r), then $\psi = \{(r, r), \ldots, (r, c)\}$ by (g2). Any $(r', c') \in \hat{\varphi}$ has $r' > r$, so $r < r' \leq c' \leq c$, and column c' is a column of ψ, a contradiction. If ψ is a column in column c containing (c, c), then $\psi = \{(r, c), \ldots, (c, c)\}$. Any $(r', c') \in \hat{\varphi}$ has $r \leq r' \leq c' \leq c$, so row r' is a row of ψ, a contradiction.
- By (g5) and the first bullet, the constraints on ψ and on $\hat{\varphi}$ are independent, so $K(\varphi) = K(\psi)K(\hat{\varphi})$. Here $K(\psi) = 2$ for $a = 1$ (an off-diagonal single box) and $K(\psi) = 1$ for $a = 2$ (an off-diagonal domino, labels $1', 1$).
- Hence $T(\varphi) = (-1)^a K(\psi) T(\hat{\varphi})$, which is $-2\mathcal{C}(\hat{\varphi}, 1)$ for $a = 1$ and $+\mathcal{C}(\hat{\varphi}, 1)$ for $a = 2$.

Every nonempty rim falls under exactly one of the cases, since $\hat{\varphi} = \varnothing$ iff φ is a row or a column. □

[CHECK: The case analysis in the proof of Proposition 4.6 has been re-read once by the author. An independent referee (Clio Vega) has verified its conclusion on 1243 shapes, with live negative controls, but has not yet read the geometric case analysis line by line. The conclusion was also checked by machine against Buch–Ravikumar's eq. (11) for all shapes arising at $m \leq 7$ (including non-rims).]

Theorem 4.7 (The Pieri coefficients)**.** *On* $\mathrm{LG}(m, 2m)$*, for every* $\rho \subsetneq \kappa$ *with* $\theta = D(\kappa) \setminus D(\rho)$*,*

$$c_\rho = (-1)^{|\theta|-1} V(\theta).$$

Explicitly, $c_\rho = 0$ *iff* θ *contains a 3-chain or a diagonal box* (c, c) *with* $(c-1, c) \in \theta$*. Otherwise* $|c_\rho| = 2^i$*, where* i *is the number of off-diagonal boxes of* θ *that are comparable to no other box of* θ*.*

Proof. Lemma 4.2 and [4, Cor. 5.4] give $c_\rho = \mathcal{C}(R(\theta), 1)$. Proposition 4.6 turns this into $(-1)^{|\theta|-1} K(R(\theta))$, and Lemma 4.5 gives $K(R(\theta)) = V(\theta)$. The explicit form is Lemma 4.3 together with the remark following it. □

Remark 4.8 (Primes and odd values). Under R, the letters $1' < 1$ of Buch and Ravikumar become $o > e$ on θ, and the rule that diagonal boxes carry no prime becomes the rule that diagonal boxes take the even letter. In the peeling below, e and o are realized by consecutive values $2j < 2j+1$. So the odd values in Theorem 1.1(b) are the primed letters, and the diagonal parity condition is the "no primed diagonal" rule.

Example 4.9. Let $\kappa = (3, 2)$, $m = 3$. Theorem 4.7 gives

$$[\mathcal{O}_{X_{(3,2)}}] \cdot \mathcal{O}^1 = 2[\mathcal{O}_{X_{(3,1)}}] - [\mathcal{O}_{X_{(2,1)}}] - [\mathcal{O}_{X_{(3)}}] + [\mathcal{O}_{X_{(2)}}].$$

The sets θ are $\{(1, 2)\}$ (an isolated off-diagonal box, $V = 2$), $\{(0, 2), (1, 2)\}$ (a 2-chain, $V = 1$), $\{(1, 1), (1, 2)\}$ (diagonal box minimal, $V = 1$) and $\{(0, 2), (1, 1), (1, 2)\}$ (height 2, $(1, 1)$ minimal, $V = 1$). The signs are $(-1)^{|\theta|-1} = +, -, -, +$. The remaining candidates $\rho = (1)$ and $\rho = \varnothing$ have $c_\rho = 0$, because θ contains the 3-chain $(0, 1) < (0, 2) < (1, 2)$. This agrees with Buch and Ravikumar's formula [4, eq. (11)] evaluated directly.

4.4. **The recursion and the second proof.**

Proposition 4.10. *On* LG*, put* $s_x(\kappa) = (-1)^{|\kappa|} H_\kappa(-x)$ *for* $x \in \mathbb{C}$*. Then for every* $x \in \mathbb{C}$*,*

$$s_{x+1}(\kappa) = s_x(\kappa) + \sum_{\rho \subsetneq \kappa} V\big(D(\kappa) \setminus D(\rho)\big)\, s_x(\rho). \tag{2}$$

The same holds on OG *with* $V(\theta)$ *replaced by* $|c_\rho|$*, where* c_ρ *are the* OG *coefficients of Lemma 4.1.*

Proof. Substitute $j = -x$ in (1) and multiply by $(-1)^{|\kappa|}$. With $|\kappa| = |\rho| + |\theta|$ this gives $s_x(\kappa) - s_{x+1}(\kappa) = \sum_\rho (-1)^{|\theta|} c_\rho s_x(\rho)$. By Theorem 4.7, $(-1)^{|\theta|} c_\rho = -V(\theta)$. On OG, [4, Lemma 4.5] gives the sign $(-1)^{|\theta|-1}$ on the support (Lemma 4.12 below). □

Lemma 4.11 (Peeling the top two values)**.** *For* $j \geq 1$ *and every* κ*,*

$$N_\kappa(j+1) = N_\kappa(j) + \sum_{\rho \subsetneq \kappa} V\big(D(\kappa) \setminus D(\rho)\big)\, N_\rho(j).$$

Proof. Let $f \colon P_\kappa \to [2j+1]$ be strictly order-preserving with even diagonal values, and put $\theta = f^{-1}\{2j, 2j+1\}$. If $x \in \theta$ and $x < y$, then $f(y) > f(x) \geq 2j$, so $y \in \theta$. Thus θ is an up-set, and $\theta = D(\kappa) \setminus D(\rho)$ for a unique ρ (Lemma 2.1). The restriction $f|_\theta$ is strictly order-preserving into $\{2j < 2j+1\}$ and sends diagonal boxes to $2j$, the only even value. So it is one of the maps counted by $V(\theta)$, with $e = 2j$ and $o = 2j+1$. The restriction $f|_{D(\rho)}$ is counted by $N_\rho(j)$. Conversely, given ρ, a $V(\theta)$-map g and an $N_\rho(j)$-map h, the union $g \cup h$ is strictly order-preserving: for $x \in D(\rho)$, $y \in \theta$ with $x < y$ we have $h(x) \leq 2j - 1 < 2j \leq g(y)$, and $y < x$ is impossible because θ is an up-set. The two constructions are inverse. The term $\rho = \kappa$ ($\theta = \varnothing$, $V = 1$) gives $N_\kappa(j)$. □

Second proof of Theorem 1.1. We prove (b) by induction on j, simultaneously for all κ. For $j = 1$, the set $[1]$ contains no even value, and every nonempty P_κ contains the diagonal box $(0,0)$. So $N_\kappa(1) = \delta_{\kappa,\varnothing}$, which equals $s_1(\kappa)$ by Lemma 2.6 (and $H_\varnothing \equiv 1$). For the inductive step, (2) at integer $x = j$ and Lemma 4.11 are the same recursion. Parts (a) and (c) then follow from (b) by running §3.4 backwards. Indeed, both H_κ and $L(j) = \#(j\mathcal{O} \cap \Lambda)$ are polynomials, and by Ehrhart–Macdonald reciprocity they agree at every $j \leq -1$. □

The second proof uses (G0), (G1) and (BR). It uses neither (LS) nor Ehrhart–Macdonald reciprocity for (b). The first proof uses (G1) and (LS) and not (BR). Each proof gives the equivalence of Theorem 1.1(a) and (b) by reciprocity, in opposite directions.

4.5. **The orthogonal Grassmannian.**

Lemma 4.12. *On* $\mathrm{OG}(m, 2m+1)$*,* $c_\rho \neq 0$ *iff* $\theta = D(\kappa) \setminus D(\rho)$ *is a nonempty set of maximal elements of* P_κ*, and then* $c_\rho = (-1)^{|\theta|-1}$*.*

Proof. By Lemma 4.2 and [4, Cor. 4.4], $c_\rho = \mathcal{B}(R(\theta), 1)$, and R preserves the properties "rim" and "at most k boxes per row and column". By [4, Lemma 4.5] with $p = 1 < a = 2$, $c_\rho \neq 0$ forces $R(\theta)$, hence θ, to be a rook strip, and every rook strip has $c_\rho = (-1)^{|\theta|-1}$. For an up-set θ of P_κ, being a rook strip is equivalent to being an antichain. Two boxes in one row or column are comparable, so an antichain is a rook strip. Conversely, if $x = (i,j) < y = (k,l)$ in θ with $i < k$ and $j < l$, then (i,l) lies in the shifted plane, below y and above x. So $(i,l) \in \theta$ (since θ is an up-set of the order ideal $D(\kappa)$, and $(i,l) \leq y$ forces $(i,l) \in D(\kappa)$), and it shares a row with x. An up-set is an antichain iff it consists of maximal elements, and every set of maximal elements is an up-set. □

K-theoretic proof of Theorem 1.3. A strictly order-preserving $f\colon P_\kappa \to [n+1]$ is determined by the set $A = f^{-1}(n+1)$, which is an up-set and an antichain, hence a set of maximal elements, together with a strictly order-preserving map $P_\kappa \setminus A \to [n]$. This gives $\overline{\Omega}(P_\kappa, n+1) = \overline{\Omega}(P_\kappa, n) + \sum_{\varnothing \neq A \subseteq \max P_\kappa} \overline{\Omega}(P_\kappa \setminus A, n)$, and $P_\kappa \setminus A = D(\rho_A)$. By Lemma 4.12, the recursion of Proposition 4.10 on OG is the same recursion with $n = j-1$. The anchor is $\overline{\Omega}(P, 0) = \delta_{P,\varnothing} = s_1$ (Lemma 2.6). So $(-1)^{|\kappa|} H^{\mathrm{OG}}_\kappa(-j) = \overline{\Omega}(P_\kappa, j-1)$ for $j \geq 1$. Stanley reciprocity $\Omega(P, 1-j) = (-1)^{|P|}\overline{\Omega}(P, j-1)$ shows that the polynomials $H^{\mathrm{OG}}_\kappa(j)$ and $\Omega(P_\kappa, j+1)$ agree at all $j \leq -1$, hence everywhere. □

Remark 4.13 (Theorem 1.3 is classical). Since ω_m is minuscule for B_m, standard monomial theory [15, 23] and the toric degeneration of Gonciulea and Lakshmibai [10] identify the homogeneous coordinate ring of X^{OG}_κ with an algebra with straightening law on the distributive lattice $J(P_\kappa)$, with the same Hilbert function as the Hibi ring [11]. Hence $H^{\mathrm{OG}}_\kappa(j) = \Omega(P_\kappa, j+1)$; compare also Proctor [21]. Stanley's reciprocity theorem then gives the negative values. We know of no single reference that states $(-1)^{|\kappa|} H^{\mathrm{OG}}_\kappa(-j) = \overline{\Omega}(P_\kappa, j-1)$ in this form, but it is a one-line composite of these results, and we claim no novelty for it. **[CHECK: the classical references in this remark (Lakshmibai–Seshadri, Gonciulea–Lakshmibai, Hibi, Proctor) were not read first-hand for this draft.]**

5. The integer window

Proof of Theorem 1.4 from the counts. On LG, $s_j(\kappa) = N_\kappa(j)$ by Theorem 1.1(b). On OG, $s_j(\kappa) = \overline{\Omega}(P_\kappa, j-1)$ by Theorem 1.3. Lemma 2.3 gives the vanishing range, the positivity beyond it, and the monotonicity. For $\kappa = \varnothing$, $s_j \equiv 1$ and $W = 0$. □

The theorem also has a direct proof by induction. That proof does not pass through any counting formula for s_j. It uses only (2), the anchor, a bound on how much r can drop along the support of the Pieri rule, and one explicit nonzero Pieri term. We include it because it shows why integers are easier than half-integers: the anchor value 0 makes the recursion (2) monotone.

Lemma 5.1 (Chain drop)**.** *If $c_\rho \neq 0$, then every chain of P_κ meets $\theta = D(\kappa) \setminus D(\rho)$ in at most two boxes on* LG*, and in at most one box on* OG*. Hence $r(\rho) \geq r(\kappa) - 2$ on* LG*, and $r(\rho) \geq r(\kappa) - 1$ on* OG*. The same bounds hold for $r(\rho')$ and $r(\kappa')$.*

Proof. On LG, θ has no 3-chain by Theorem 4.7 and Lemma 4.3. On OG, θ is an antichain by Lemma 4.12. If C is a longest chain of P_κ, then $C \setminus \theta \subseteq D(\rho)$ is a chain of the claimed size. For the primed version, take C inside the rows ≥ 1, which form copies of $P_{\kappa'}$ and $P_{\rho'}$ (§2.1). Row 0 is the largest part of both κ and ρ. □

Lemma 5.2 (A term that drops the window)**.** *Let $\kappa \neq \varnothing$, $r = r(\kappa)$, and for $t \in \{1, 2\}$ put $T_t = \{(a, b) \in D(\kappa) : a + b \geq r - t\}$ and $D(\rho_t) = D(\kappa) \setminus T_t$.*

(i) ρ_t is a strict partition with $\rho_t \subsetneq \kappa$. If $t \leq r$, then $r(\rho_t) \leq r - t$.
(ii) On LG*, let $t^* = 1$ if r is odd and $t^* = 2$ if r is even. Then $c_{\rho_{t^*}} \neq 0$.*
(iii) On OG*, $c_{\rho_1} = (-1)^{|T_1|-1}$.*
(iv) $W(\rho_{t^*}) \leq W(\kappa) - 1$ on LG*, and $W(\rho_1) \leq W(\kappa) - 1$ on* OG*.*

Proof. (i) T_t is an up-set of P_κ, so ρ_t exists by Lemma 2.1. It contains the box realising $\max(a+b)$ (Lemma 2.2), so $\rho_t \neq \kappa$. If $t \leq r$ and $\rho_t \neq \varnothing$, every box of $D(\rho_t)$ has $a + b \leq r - t - 1$, so $r(\rho_t) \leq r - t$ by Lemma 2.2. If $\rho_t = \varnothing$, then $r(\rho_t) = 0 \leq r - t$. (The hypothesis $t \leq r$ is needed: for $\kappa = (1)$ and $t = 2$ the bound would read $0 \leq -1$. We only use $t = t^* \leq r$.)

(ii) By Theorem 4.7 and Lemma 4.3 it suffices to show that T_{t^*} has no 3-chain and that each of its diagonal boxes is minimal in it. The coordinate sum strictly increases along a chain, and T_{t^*} lies on at most two antidiagonals $a + b = \text{const}$, so there is no 3-chain. A box of T_{t^*} strictly

below (c,c) has a smaller coordinate sum. For r odd, T_1 is a single antidiagonal, so no such box exists. For r even, $T_2 = \{a+b \in \{r-2, r-1\}\}$, and $2c$ is even, so $2c = r-2$ lies on the inner antidiagonal; again nothing in T_2 lies below (c,c).

(iii) T_1 is the top antidiagonal: it is nonempty, and it consists of maximal elements of P_κ. Apply Lemma 4.12.

(iv) On LG with r odd, $r(\rho_1) \le r-1$, which is even, so $\lceil r(\rho_1)/2 \rceil \le (r-1)/2 = W-1$. With r even, $r(\rho_2) \le r-2$, so $\lceil r(\rho_2)/2 \rceil \le r/2 - 1 = W-1$. On OG, $r(\rho_1) \le r-1 = W-1$. □

The parity choice in (ii) matters. For r odd the two-strip T_2 has its diagonal box (if any) on the *outer* antidiagonal, where it can lie above $(c-1,c) \in T_2$; then $V(T_2) = 0$. A machine check of the 67 odd-r two-strips at $m \le 6$ against Buch–Ravikumar's formula finds that 42 of them have vanishing coefficient, each one exactly in this situation.

Inductive proof of Theorem 1.4. We prove by strong induction on $|\kappa|$ that $s_j(\kappa) \ge 0$ for all integers $j \ge 1$, with equality iff $j \le W(\kappa)$. For $\kappa = \varnothing$ this is clear. Let $\kappa \ne \varnothing$.

Monotonicity. For $j \ge 1$, every ρ in (2) has $|\rho| < |\kappa|$, so $s_j(\rho) \ge 0$ by induction, and the coefficients in (2) are nonnegative. Hence $s_{j+1}(\kappa) \ge s_j(\kappa)$.

Anchor. $s_1(\kappa) = 0$ by Lemma 2.6. So $s_j(\kappa) \ge 0$ for all $j \ge 1$.

Vanishing. Let $1 \le j \le W-1$. For every ρ with $c_\rho \ne 0$, Lemma 5.1 gives $W(\rho) \ge W-1 \ge j$: on LG, $\lceil r(\rho)/2 \rceil \ge \lceil (r-2)/2 \rceil = W-1$, and on OG, $r(\rho) \ge r-1$. So $s_j(\rho) = 0$ by induction, and (2) gives $s_{j+1}(\kappa) = s_j(\kappa)$. Starting from $s_1 = 0$, this reaches $s_W = 0$.

Nonvanishing. At $j = W$, (2) gives $s_{W+1}(\kappa) \ge |c_{\rho^*}|\, s_W(\rho^*)$ with $\rho^* = \rho_{t^*}$ (on LG) or ρ_1 (on OG); the other terms are ≥ 0. By Lemma 5.2, $c_{\rho^*} \ne 0$ and $W(\rho^*) < W$, so $s_W(\rho^*) > 0$ by induction. Monotonicity gives $s_j > 0$ for all $j > W$. □

Example 5.3. For $\kappa = (3,2)$ on LG$(3,6)$: the coordinate sums of $D(\kappa)$ are $0,1,2,2,3$, so $r = 4$ and $W = 2$. By Example 4.9, (2) at $j = 2$ reads $s_3(3,2) = s_2(3,2) + 2s_2(3,1) + s_2(2,1) + s_2(3) + s_2(2)$. The first four terms vanish, since each of those κ has $W = 2$, and $s_2(2) = 1$. So $s_3(3,2) = 1$, that is, $H_{(3,2)}(-3) = -1$. Here $\rho^* = \rho_2 = (2)$ and $T_2 = \{(0,2),(1,1),(1,2)\}$. At $j = 3$ the same recursion gives $s_4(3,2) = 1 + 2\cdot 2 + 1 + 3 + 4 = 13$. For $\kappa = (4,3,2,1)$ we have $r = 7$ and $W = 4$; the support of $[\mathcal{O}_{X_\kappa}] \cdot \mathcal{O}^1$ is the single term $\rho = (4,3,2)$ with $\theta = T_1 = \{(3,3)\}$ and $c_\rho = 1$.

6. Half-integer values

Throughout this section $X = \mathrm{LG}(m,2m)$, and $s_x(\kappa) = (-1)^{|\kappa|} H_\kappa(-x)$ for $x \in \frac{1}{2}\mathbb{Z}$. Since (2) is an identity of polynomials, it holds at half-integers as well. Write $y_i = e^{-\varepsilon_i}$.

6.1. The anchor at $-\frac{1}{2}$ and the bridge to vector bundles.

For $1 \le k \le m$, put $c(k) = s_{m-k+1} s_{m-k+2} \cdots s_m$, of length k. Then $D_{c(k)} = D_{m-k+1} \circ \cdots \circ D_m$, so D_m acts first.

Lemma 6.1 (Reduced word)**.** *$w_\kappa = c(\kappa_\ell) \cdots c(\kappa_2)\, c(\kappa_1)$, and this word is reduced. Equivalently, $w_\kappa = w_{\kappa'}\, c(\kappa_1)$ with lengths adding, and every letter of $w_{\kappa'}$ is s_j with $j \ge m - \kappa_1 + 2$.*

Proof. Let $p = m - \kappa_1 + 1$. Applying the letters of $c(\kappa_1)$ to $\omega = (1,\dots,1)$ from right to left, s_m flips the last coordinate and $s_{m-1}, \dots, s_p$ carry the -1 to position p. So $c(\kappa_1)\omega = v_{(\kappa_1)}$. Every letter of $c(\kappa_\ell) \cdots c(\kappa_2)$ is s_j with $j \ge m - \kappa_2 + 1 \ge p+1$, so it only moves coordinates $\ge p+1$, where $v_{(\kappa_1)}$ is $+1$. By induction on $\ell(\kappa)$, the word maps ω to v_κ. It has length $|\kappa| = \ell(w_\kappa)$ and lies in the coset $w_\kappa W_P$, and the minimal coset representative is the unique element of minimal length. So the word is reduced and equals w_κ. □

Let $D^B_m g = (g - y_m s_m g)/(1 - y_m)$ be the type-B operator for the short root ε_m. Put $D'_i = D_i$ for $i < m$ and $D'_m = (1+y_m)^{-1} D^B_m$, and let $\Phi = \prod_{i=1}^m (1+y_i)^{-1}$, an analytic germ with $\Phi(0) = 2^{-m}$.

Lemma 6.2 (Conjugation). *For all i, $D_i(e^{-\omega/2}g) = e^{-\omega/2}D_i'(g)$ and $D_i'(\Phi g) = \Phi D_i(g)$. Hence $D_w(e^{-\omega/2}g) = e^{-\omega/2}\,\Phi\, D_w(\Phi^{-1}g)$ for every w.*

Proof. For $i < m$, both $e^{-\omega/2}$ and Φ are s_i-invariant, and $D_i(\varphi g) = \varphi D_i g$ whenever $s_i\varphi = \varphi$. For $i = m$: $s_m(e^{-\omega/2}) = e^{-\omega/2}e^{\varepsilon_m}$ and $e^{-\alpha_m} = y_m^2$, so $D_m(e^{-\omega/2}g) = e^{-\omega/2}(g - y_m s_m g)/(1-y_m^2) = e^{-\omega/2}(1+y_m)^{-1}D_m^B g$. For the second identity, write $\Phi = \Phi'(1+y_m)^{-1}$ with Φ' s_m-invariant, and use $s_m(1+y_m)^{-1} = y_m(1+y_m)^{-1}$. Then $D_m^B((1+y_m)^{-1}g) = (g - y_m^2 s_m g)/((1+y_m)(1-y_m))$, and multiplying by $(1+y_m)^{-1}\Phi'$ gives $\Phi(g - y_m^2 s_m g)/(1-y_m^2) = \Phi D_m g$. □

Proposition 6.3 (Bridge). *For every integer $a \geq 0$, $2^m H_\kappa(-a-\frac{1}{2}) = \left[D_{w_\kappa}(F_a)\right]_{y=1}$, where $F_a = \prod_{i=1}^m y_i^a(1+y_i)$.*

Proof. By Lemma 2.5,

$$H_\kappa(-a-\tfrac{1}{2}) = [D_{w_\kappa}(e^{-\omega/2}e^{-a\omega})](0).$$

By Lemma 6.2 this equals $[e^{-\omega/2}\Phi\, D_{w_\kappa}(F_a)](0)$, because $\Phi^{-1}e^{-a\omega} = F_a$, and $e^{-\omega/2}\Phi = 2^{-m}$ at 0. □

Remark 6.4. F_a is the character of $\mathbb{C}_{-a\omega} \otimes \bigwedge^\bullet U_0$, where U_0 is the P-module with weights $-\varepsilon_i$, the fibre of the tautological subbundle U. Pulling back to G/B and filtering by B-weights, (G1) turns Proposition 6.3 into $2^m H_\kappa(-a-\frac{1}{2}) = \chi(X_\kappa, \bigwedge^\bullet U \otimes \mathcal{O}(-a))$. So the half-integer values are Euler characteristics of honest vector bundles. (For $m=1$, $X = \mathbb{P}^1$ and $U = \mathcal{O}(-1)$: $\chi(\mathcal{O}\oplus\mathcal{O}(-1)) = 1 = 2H(-\frac{1}{2})$.) None of the proofs below use this geometric reading.

Theorem 6.5. *$H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$ for every κ (with $\kappa_1 = 0$ for $\kappa = \varnothing$).*

Proof. By Proposition 6.3 with $a=0$, $2^m H_\kappa(-\frac{1}{2}) = [D_{w_\kappa}\prod_i(1+y_i)]_{y=1}$. Let $p = m - \kappa_1 + 1$. We claim that $D_{c(\kappa_1)}\prod_{i=1}^m(1+y_i) = \prod_{i<p}(1+y_i)$. Apply $D_m, D_{m-1}, \ldots, D_p$ in turn. Before D_q acts, the function is $\prod_{i\leq q}(1+y_i)$ (true for $q=m$), and $\prod_{i<q}(1+y_i)$ is s_q-invariant. So $D_q\prod_{i\leq q}(1+y_i) = \prod_{i<q}(1+y_i)\,(1+D_q e^{-\varepsilon_q})$. Here $\langle -\varepsilon_q, \alpha_q^\vee\rangle = -1$ for $q < m$ and for $q = m$, so $D_q e^{-\varepsilon_q} = 0$. By Lemma 6.1, the remaining letters s_j have $j \geq p+1$; they fix $\prod_{i<p}(1+y_i)$, and $D_j(1) = 1$. Evaluating at $y=1$ gives $2^{p-1} = 2^{m-\kappa_1}$. □

Theorem 6.6. *If $\kappa_1 \geq 2$, then $H_\kappa(-\frac{3}{2}) = 2^{-\kappa_1}H_{\kappa'}(-1)$, and $H_{(1)}(-\frac{3}{2}) = -\frac{1}{2}$. Hence $H_\kappa(-\frac{3}{2}) = 0$ iff $\ell(\kappa) \geq 2$.*

Proof. For $\kappa = (1)$, $H_{(1)}(j) = j+1$. Let $\kappa_1 \geq 2$, $p = m-\kappa_1+1$, and $f(y) = y+y^2$, so $F_1 = \prod_i f(y_i)$. We apply $D_{c(\kappa_1)}$. D_m kills y_m (pairing -1) and sends y_m^2 to -1 (pairing -2), so $D_m F_1 = -\prod_{i<m} f(y_i)$. Next, D_{m-1} kills y_{m-1} and sends y_{m-1}^2 to $-y_{m-1}y_m$, so $D_{m-1}(-f(y_{m-1})) = y_{m-1}y_m$. For $p \leq q < m-1$, D_q applied to $f(y_q)y_{q+1}\cdots y_m$ fixes $y_q y_{q+1}\cdots y_m$ (pairing 0) and kills $y_q^2 y_{q+1}\cdots y_m$ (pairing -1). Hence $D_{c(\kappa_1)}F_1 = \prod_{i<p} f(y_i)\cdot y_p\cdot(y_{p+1}\cdots y_m)$. By Lemma 6.1, $D_{w_{\kappa'}}$ acts only on the coordinates $p+1,\ldots,m$, where $y_{p+1}\cdots y_m = e^{-\omega''}$ for the rank-(κ_1-1) block. The prefactor is inert and equals 2^{p-1} at $y=1$. The Demazure expression for $H_{\kappa'}$ does not depend on the rank, by the same inert-factor argument. Hence $2^m H_\kappa(-\frac{3}{2}) = 2^{m-\kappa_1}H_{\kappa'}(-1)$. Finally, $H_{\kappa'}(-1) = 0$ iff $\kappa' \neq \varnothing$ (Lemma 2.6), and $H_\varnothing \equiv 1$. □

6.2. **Vanishing by hyperplane sections.**

Theorem 6.7. *For every strict κ and every odd M with $3 \leq M \leq 2\lceil r(\kappa')/2\rceil + 1$, $H_\kappa(-M/2) = 0$.*

Proof. Strong induction on $|\kappa|$. If $\ell(\kappa) \leq 1$, then $r(\kappa') = 0$ and the range is empty. Let $\ell(\kappa) \geq 2$, so $r' = r(\kappa') \geq 1$. Put $b_\kappa(x) = H_\kappa(x) - H_\kappa(x-1)$.

Every term vanishes on the shrunken window. Let M be odd with $3 \le M \le 2\lceil r'/2\rceil - 1$, and let $c_\rho \neq 0$. By Lemma 5.1, $r(\rho') \ge r' - 2$, so $2\lceil r(\rho')/2\rceil + 1 \ge 2\lceil (r'-2)/2\rceil + 1 = 2\lceil r'/2\rceil - 1 \ge M$. Thus M lies in the window of ρ, and $H_\rho(-M/2) = 0$ by induction. By (1), $b_\kappa(-M/2) = 0$.

Telescoping from the anchor. By Theorem 6.6, $H_\kappa(-\frac{3}{2}) = 0$. For M in the range above, $b_\kappa(-M/2) = 0$ says $H_\kappa(-M/2) = H_\kappa(-(M+2)/2)$. Going up $M = 3, 5, \dots, 2\lceil r'/2\rceil - 1$ carries the zero from $-\frac{3}{2}$ to $-(2\lceil r'/2\rceil + 1)/2$. □

Theorem 6.7 uses only the support of the Pieri rule: it needs to know that $c_\rho \neq 0$ implies that θ has no 3-chain. This also follows from the rim condition of [4, Thm. 5.5] together with the bound of two boxes per row and column (two boxes in strictly south-east position, or three in a row or column, would be needed for a 3-chain). The coefficients do not cooperate at the edge of the window: the nonzero terms of b_κ at $-M^*/2$, $M^* = 2\lceil r'/2\rceil + 1$, need not have a common sign (for $\kappa = (3,2)$ they are $-\frac{1}{8}$ and $+\frac{1}{4}$). So the nonvanishing half needs the finer argument below.

6.3. **The odd-zero law.** Let A_o and A_e be the matrices indexed by strict partitions with $A_o(\kappa,\nu) = [\theta \text{ is an antichain without diagonal boxes}]$ and $A_e(\kappa,\nu) = [\theta \text{ is an antichain}]$, for $\nu \subseteq \kappa$ and $\theta = D(\kappa) \setminus D(\nu)$, and 0 otherwise.

Lemma 6.8 (Layers). *(a)* $(A_oA_e)(\kappa,\nu) = V(D(\kappa) \setminus D(\nu))$. *Hence* (2) *reads* $s_{x+1} = A_oA_e\, s_x$ *for every* $x \in \mathbb{C}$.
(b) For $a \ge 1$, $[(A_oA_e)^{a-1}A_o](\kappa,\nu) = N_{\kappa/\nu}(a)$.

Proof. Peel values from the top, as in Lemma 4.11. The preimage of the current top value is an antichain that is an up-set of what remains. The value is odd (no diagonal boxes allowed: A_o) or even (A_e). In (b), the values $2a-1, 2a-2, \dots, 1$ alternate $o, e, \dots, o$, giving $A_o(A_eA_o)^{a-1} = (A_oA_e)^{a-1}A_o$. Part (a) is the same with the alphabet $\{e < o\}$, and $\theta = \varnothing$ gives 1. □

Lemma 6.9 (The anchor collapses to single rows). *Let* $\varepsilon = A_e\, s_{1/2}$. *Then* $\varepsilon(\varnothing) = 1$, $\varepsilon((L)) = (-1)^{L+1}2^{-L}$ *for* $L \ge 1$, *and* $\varepsilon(\kappa) = 0$ *if* $\ell(\kappa) \ge 2$.

Proof. By Theorem 6.5, $s_{1/2}(\nu) = (-1)^{|\nu|}2^{-\nu_1}$, and $\varepsilon(\kappa) = \sum_{A \subseteq \max P_\kappa} s_{1/2}(\kappa \setminus A)$. If $\ell(\kappa) \ge 2$, the last box z of the bottom row is maximal and not in row 0. Toggling $z \in A$ changes $|\nu|$ by one and leaves ν_1 alone, so the terms cancel in pairs. If $\kappa = (L)$, the only maximal box is $(0, L-1)$, and $\varepsilon = (-1)^L 2^{-L} + (-1)^{L-1}2^{-(L-1)} = (-1)^{L+1}2^{-L}$. □

Corollary 6.10 (Row formula). *For* $a \ge 1$, $s_{a+1/2}(\kappa) = N_\kappa(a) + \sum_{L=1}^{\kappa_1} (-1)^{L+1}2^{-L} N_{\kappa/(L)}(a)$.

Proof. $s_{a+1/2} = (A_oA_e)^a s_{1/2} = (A_oA_e)^{a-1}A_o\,\varepsilon$; apply Lemmas 6.8(b) and 6.9. □

For example, for $\kappa = (3)$ and $a = 2$: $s_{5/2}(3) = 0 + \frac{1}{2}\cdot 3 - \frac{1}{4}\cdot 3 + \frac{1}{8}\cdot 1 = \frac{7}{8}$, and indeed $H_{(3)}(-\frac{5}{2}) = -\frac{7}{8}$.

The fibre over the lower rows. Let $n = \kappa_1$, $M = 2a-1$, and $k = \kappa_2$ (with $k = 0$ if $\ell(\kappa) = 1$). Rows ≥ 1 of P_κ form a copy of $P_{\kappa'}$. Row 0 is the chain $(0,0) < \cdots < (0, n-1)$, and its cover relations to rows ≥ 1 are $(0,\beta) < (1,\beta)$ for $1 \le \beta \le k$; nothing in rows ≥ 1 lies below a box of row 0. Fix a strictly order-preserving g on rows ≥ 1 with values in $[M]$ and even diagonal values; there are $N_{\kappa'}(a)$ of them. Put $u_\beta = g(1,\beta)$ for $1 \le \beta \le k$ and $u_\beta = M+1$ (no bound) for $k < \beta \le n-1$. For a bound vector u and $1 \le L \le n$, let $G_L(M;u)$ be the number of strictly increasing $v_L < \cdots < v_{n-1}$ in $[M]$ with $v_\beta < u_\beta$ (so $G_n = 1$), and let $E_0(M;u)$ be the same count for $L = 0$ with v_0 even. Put

$$\Phi(u) = E_0(M;u) - \sum_{L=1}^{n} (-\tfrac{1}{2})^L G_L(M;u).$$

Summing Corollary 6.10 over g gives

$$s_{a+1/2}(\kappa) = \sum_g \Phi(u(g)). \tag{3}$$

The bounds coming from an actual g satisfy

$$u_\beta \geq \beta + 1 \quad \text{for } 1 \leq \beta \leq k, \tag{H}$$

because $u_1 = g(1,1)$ is even, hence at least 2, and row 1 increases strictly.

Let $\Psi_L(M;u) = \sum_{L'=L}^{n} (-\frac{1}{2})^{L'-L} G_{L'}(M;u)$. Then $\Psi_n = 1$ and $\Phi = E_0 + \frac{1}{2}\Psi_1$.

Lemma 6.11 (Shift)**.** *Let $1 \leq L \leq n-1$ and $M \geq 1$, and suppose $u_{L'} \geq 2$ for $L \leq L' \leq n-1$. Then $\Psi_L(M;u) = G_L(M-1;u-1) + \frac{1}{2}\Psi_{L+1}(M-1;u-1)$.*

Proof. For $L' \geq L$, let $\alpha_{L'}$ (resp. $\beta_{L'}$) count the fillings of boxes $L', \ldots, n-1$ with $v_{L'} = 1$ (resp. $v_{L'} \geq 2$), and put $\beta_n = 1$. Then $G_{L'} = \alpha_{L'} + \beta_{L'}$ for $L' \leq n-1$. Deleting box L' (value 1) is a bijection onto the fillings of $L'+1, \ldots, n-1$ with all values ≥ 2. Its inverse puts 1 in box L', which is allowed because $u_{L'} \geq 2$ and $M \geq 1$. So $\alpha_{L'} = \beta_{L'+1}$ for $L \leq L' \leq n-1$. Substituting,

$$\Psi_L = \sum_{L'=L}^{n-1} (-\tfrac{1}{2})^{L'-L}(\beta_{L'} + \beta_{L'+1}) + (-\tfrac{1}{2})^{n-L}\beta_n = \beta_L + \tfrac{1}{2}\sum_{L''>L} (-\tfrac{1}{2})^{L''-L-1}\beta_{L''}.$$

Finally, $v \mapsto v-1$ identifies fillings with values in $[2,M]$ under bounds u with fillings in $[M-1]$ under bounds $u-1$, so $\beta_{L''}(M;u) = G_{L''}(M-1;u-1)$, including $L'' = n$. □

Lemma 6.12 (One row)**.** *Assume (H).*

(a) If $M \geq n-1$, then $\Phi(u) = E_0(M;u) + \sum_{i=0}^{n-2} 2^{-i-1} G_{1+i}(M-1-i;u-1-i) + 2^{-n} \geq 2^{-n}$.
(b) If $M \leq n-1$, then $\Phi(u) = (-1)^{n-1-M}2^{-n}$.

Proof. Apply Lemma 6.11 repeatedly. Step $i = 0, 1, \ldots$ rewrites $\Psi_{1+i}(M-i;u-i)$. Its hypotheses are $M - i \geq 1$ and $u_{L'} - i \geq 2$ for $L' \geq 1+i$. The second holds by (H) for bounded boxes ($u_{L'} - i \geq L' + 1 - i \geq 2$), and for unbounded boxes as long as $M - i \geq 1$. (a) If $M \geq n-1$, steps $i = 0, \ldots, n-2$ are allowed, ending at $\Psi_n = 1$, so $\Psi_1 = \sum_{i=0}^{n-2} 2^{-i} G_{1+i}(M-1-i;u-1-i) + 2^{-(n-1)}$. Every term is nonnegative. (b) If $M \leq n-1$, steps $i = 0, \ldots, M-1$ are allowed, ending at $\Psi_{1+M}(0;u-M) = (-\frac{1}{2})^{n-1-M}$, since only $G_n = 1$ survives an empty alphabet. The G-terms vanish: $G_{1+i}(M-1-i)$ fills $n-1-i$ boxes from $M-1-i < n-1-i$ values. So $\Psi_1 = 2^{-M}(-\frac{1}{2})^{n-1-M}$. Also $E_0 = 0$, because n strictly increasing values that are all ≥ 2 need $M \geq n+1$. Hence $\Phi = \frac{1}{2}\Psi_1$. □

Proof of Theorem 1.5. The value at $-\frac{1}{2}$ is Theorem 6.5. Let $a \geq 1$ and $M = 2a-1$. If $2a \leq \kappa_1$, then $M \leq n-1$, and Lemma 6.12(b) gives $\Phi \equiv (-1)^{n-2a}2^{-n} = (-1)^{\kappa_1}2^{-\kappa_1}$ for every g. So $s_{a+1/2}(\kappa) = (-1)^{\kappa_1}2^{-\kappa_1}N_{\kappa'}(a)$ by (3). If $2a \geq \kappa_1$, then $M \geq n-1$, and Lemma 6.12(a) gives $\Phi \geq 2^{-n}$ for every g, so $s_{a+1/2}(\kappa) \geq 2^{-\kappa_1}N_{\kappa'}(a)$. (For $2a = \kappa_1$ both apply, and they agree because κ_1 is even.) In both cases $s_{a+1/2}(\kappa) = 0$ iff $N_{\kappa'}(a) = 0$. By Lemma 2.3, this happens iff $\kappa' \neq \varnothing$ and $a \leq \lceil r(\kappa')/2 \rceil$. The signs follow from $H_\kappa = (-1)^{|\kappa|}s$. Each $\Phi(u)$ lies in $2^{-n}\mathbb{Z}$, so $2^{\kappa_1}s_{a+1/2}(\kappa) \in \mathbb{Z}$. □

Example 6.13. Take $\kappa = (3,2)$ and $a = 2$, so $M = 3 \geq n-1 = 2$. On rows ≥ 1, the chain $(1,1) < (1,2)$ with $(1,1)$ even and values in $[3]$ admits only $g = (2,3)$, so $u = (2,3)$. Then $E_0 = 0$ (no $v_1 < 2$ fits above an even v_0) and $G_1(2;(1,2)) = 0$. Lemma 6.12(a) gives $\Phi = \frac{1}{8}$, so $s_{5/2}(3,2) = \frac{1}{8} = 2^{-\kappa_1}N_{(2)}(2)$, and $H_{(3,2)}(-\frac{5}{2}) = -\frac{1}{8}$. The bound in Theorem 1.5(iii) is attained.

Remark 6.14 (Checks). The values $(-1)^{|\kappa|}H_\kappa(-a-\frac{1}{2})$, computed from type-$C$ Demazure characters and interpolated to $x=-a-\frac{1}{2}$, agree with $\sum_g \Phi(u(g))$ for all κ with parts at most $m\le 4$ and $a\le 5$ (125 cases). Lemma 6.12 holds pointwise in all 477 cases that occur. Hypothesis (H) is necessary: without it Φ can be negative (for example $u=(1)$ gives $-\frac{1}{4}$).

6.4. **Nonvanishing for large half-integers.**

Proof of Corollary 1.6. Let $M=2a+1\ge 2m+1$, so $a\ge m\ge\kappa_1$ and $2a\ge\kappa_1$. The parts of κ' are at most $m-1$, so $\max(a'+b')\le 2m-4$ on $D(\kappa')$, and $r(\kappa')\le 2m-3$ by Lemma 2.2. Hence $\lceil r(\kappa')/2\rceil\le m-1<a$, and Theorem 1.5 gives $s_{a+1/2}(\kappa)\ge 2^{-\kappa_1}N_{\kappa'}(a)>0$ with $2^{\kappa_1}s_{a+1/2}(\kappa)\in\mathbb{Z}$. Then $2^m s_{a+1/2}(\kappa)$ is an integer, and it is at least $2^{m-\kappa_1}\ge 1$. □

Remark 6.15 (A cohomological formula). Corollary 1.6 also follows from a positive cohomological formula. That argument uses more geometry: normality and the Cohen–Macaulay property of Schubert varieties, Kodaira-type vanishing via Frobenius splitting, and the formula $\omega_Y=\mathcal{O}_Y(-\partial Y)\otimes L_{-\rho}$ for Schubert varieties $Y\subseteq G/B$ [3, Ch. 3]. Let $\pi\colon Y\to X_\kappa$ be the P/B-bundle. Serre duality applied to Remark 6.4, followed by a B-stable filtration of $\pi^*(\omega_{X_\kappa}\otimes\bigwedge^\bullet U^\vee(a))$, gives, for $a\ge m$,

$$(-1)^{|\kappa|}2^m H_\kappa(-a-\tfrac{1}{2})=\sum_{S\subseteq[m]} h^0(Y,\mathcal{O}_Y(-\partial Y)\otimes L_{\lambda_S}),\qquad \lambda_S=(a-i+[i\in S])_{i=1}^m.$$

Every λ_S is dominant for $a\ge m$, and the term $S=[m]$ is nonzero. We record this as a remark because the proof given above needs only (G0), (G1) and (BR).

7. Examples

7.1. **The quadric** $\mathrm{LG}(2,4)$ **and** $\mathbb{P}^3=\mathrm{OG}(2,5)$**.** Let $\kappa=(2,1)$, so P_κ is the chain $(0,0)<(0,1)<(1,1)$, with both ends diagonal.

Positive degrees. For $j=1$, the order-preserving maps $g\colon P_\kappa\to\{0,1,2\}$ that are even at both ends are $(0,0,0)$, $(0,0,2)$, $(0,1,2)$, $(0,2,2)$ and $(2,2,2)$. This gives $H(1)=5=\dim V_{\mathrm{Sp}_4}(\omega_2)$, the number of Plücker coordinates of the quadric $\mathrm{LG}(2,4)\subseteq\mathbb{P}^4$.

Negative integers. $H(j)=(j+1)(j+2)(2j+3)/6$ gives $H(-1)=H(-2)=0$, $H(-3)=-1$ and $H(-4)=-5$, matching Example 1.2. Here $r=3$ and $W=2$. By Serre duality with $\omega_{Q^3}=\mathcal{O}(-3)$, $H(-3)=-h^0(\mathcal{O})$.

Half-integers. $H(-\frac{1}{2})=\frac{1}{4}=2^{-\kappa_1}$, and $H(-\frac{3}{2})=0$ because $\ell(\kappa)=2$ (Theorem 6.6). Next, $H(-\frac{5}{2})=-\frac{1}{4}$: here $\kappa'=(1)$ and $r(\kappa')=1$, so the half-integer window is $a\le 1$. Since $2a=4\ge\kappa_1$, the sign is $(-1)^{|\kappa|}=-1$, and $|H(-\frac{5}{2})|=2^{-2}N_{(1)}(2)=\frac{1}{4}$.

The orthogonal twin. For $\mathrm{OG}(2,5)\cong\mathbb{P}^3$ with the spinor bundle $\mathcal{O}_{\mathbb{P}^3}(1)$, $H^{\mathrm{OG}}(j)=\Omega(P_{(2,1)},j+1)=\binom{j+3}{3}$. Its zeros are $-1,-2,-3$, so $W=r=3$, and $H^{\mathrm{OG}}(-4)=-1=-\overline{\Omega}(P_{(2,1)},3)$. The window on OG is twice as long, as Theorem 1.4 predicts.

7.2. **All Schubert varieties of** $\mathrm{LG}(3,6)$**.** Table 1 lists H_κ for all $\kappa\neq\varnothing$ with parts at most 3, together with $r=r(\kappa)$ and $r'=r(\kappa')$. Table 2 lists the values at $j=-1,\dots,-5$ and at $j=-\frac{1}{2},\dots,-\frac{9}{2}$. The polynomials were computed from Demazure characters (G1) by interpolation, independently of the theorems. The integer values agree with the strict counts of Theorem 1.1(b) for $j\le 4$. The zeros are exactly those predicted by Theorems 1.4 and 1.5: integer zeros $-1,\dots,-\lceil r/2\rceil$, and half-integer zeros $-a-\frac{1}{2}$ for $1\le a\le\lceil r'/2\rceil$ when $\kappa'\neq\varnothing$.

A few entries, read through the theorems:

- (3): $s_3(3)=3$ counts the triples $2\le x<y<z\le 5$ with x even, namely $x=2$ and $\{y,z\}\subseteq\{3,4,5\}$. At $-\frac{3}{2}$ we have $2a=2\le\kappa_1=3$, so $H=2^{-3}N_\varnothing=\frac{1}{8}$ with sign $(-1)^{|\kappa|-\kappa_1}=+$.

κ	r	r'	$H_\kappa(j)$
(1)	1	0	$j+1$
(2)	2	0	$(j+1)^2$
(3)	3	0	$(j+1)(j+2)(4j+3)/6$
$(2,1)$	3	1	$(j+1)(j+2)(2j+3)/6$
$(3,1)$	3	1	$(j+1)^2(j+2)(2j+3)/6$
$(3,2)$	4	2	$(j+1)(j+2)(2j+3)(2j^2+6j+5)/30$
$(3,2,1)$	5	3	$(j+1)(j+2)^2(j+3)(2j+3)(2j+5)/180$

TABLE 1. Hilbert polynomials of the Schubert varieties of $\mathrm{LG}(3,6)$.

κ	-1	-2	-3	-4	-5	$-\frac{1}{2}$	$-\frac{3}{2}$	$-\frac{5}{2}$	$-\frac{7}{2}$	$-\frac{9}{2}$
(1)	0	-1	-2	-3	-4	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{3}{2}$	$-\frac{5}{2}$	$-\frac{7}{2}$
(2)	0	1	4	9	16	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{9}{4}$	$\frac{25}{4}$	$\frac{49}{4}$
(3)	0	0	-3	-13	-34	$\frac{1}{8}$	$\frac{1}{8}$	$-\frac{7}{8}$	$-\frac{55}{8}$	$-\frac{175}{8}$
$(2,1)$	0	0	-1	-5	-14	$\frac{1}{4}$	0	$-\frac{1}{4}$	$-\frac{5}{2}$	$-\frac{35}{4}$
$(3,1)$	0	0	2	15	56	$\frac{1}{8}$	0	$\frac{3}{8}$	$\frac{25}{4}$	$\frac{245}{8}$
$(3,2)$	0	0	-1	-13	-70	$\frac{1}{8}$	0	$-\frac{1}{8}$	$-\frac{17}{4}$	$-\frac{259}{8}$
$(3,2,1)$	0	0	0	1	14	$\frac{1}{8}$	0	0	$\frac{1}{8}$	$\frac{35}{8}$

TABLE 2. Values of H_κ at negative integers and half-integers on $\mathrm{LG}(3,6)$.

- $(3,1)$: $H(-\frac{5}{2}) = \frac{3}{8} > 0$. Here $2a = 4 \geq 3$, so the sign is $(-1)^{|\kappa|} = +$, and the lower bound $2^{-3}N_{(1)}(2) = \frac{1}{8}$ is not attained.
- $(3,2,1) = \mathrm{LG}(3,6)$: $r = 5$, so the integer zeros are $-1, -2, -3$. Since $\kappa' = (2,1)$ has $r' = 3$, the half-integer zeros are $-\frac{3}{2}$ and $-\frac{5}{2}$. Then $H(-4) = 1$ and $H(-\frac{7}{2}) = \frac{1}{8} = 2^{-3}N_{(2,1)}(3)$.

7.3. **Computer checks.** Apart from the checks recorded with each section, Theorem 1.1(b) and Theorem 1.3 were compared with Hilbert polynomials computed directly from Demazure characters at negative weights, for all κ with parts at most m: $16, 32, 64, 128$ pairs (κ, j) with $j \leq 4$ for $m = 2, 3, 4, 5$, on both LG and OG, with no failures. Negative controls fail as they should (alphabet $[2j]$ instead of $[2j-1]$; no diagonal parity; OG alphabet $[j]$). An independent census by Clio Vega, who evaluated the Demazure characters directly at negative weights, confirmed Theorem 1.4 for all strict κ with parts at most 6, on both LG and OG.

8. Open questions

Question 8.1 (Other cominuscule spaces)**.** The cominuscule G/P that are not minuscule are $\mathrm{LG}(m, 2m)$ and the odd quadrics $Q^{2m-1} = B_m/P_1$. For the odd quadric the Bruhat poset is a chain, and exactly one cover has bond 2. This is the lecture-hall setting of [6]. Is the Hilbert function of every Schubert variety in a cominuscule G/P the Ehrhart polynomial of its order polytope with respect to the lattice determined by the bonds, as in Theorem 1.1(c)? Is there a uniform reason, independent of the case-by-case bond computation of Lemma 3.1?

Question 8.2 (Canonical modules)**.** For $j \geq 1$, Kodaira-type vanishing on Schubert varieties gives $(-1)^{|\kappa|}H_\kappa(-j) = h^0(X_\kappa, \omega_{X_\kappa}(j))$. Is there a standard-monomial-type basis of $H^0(X_\kappa, \omega_{X_\kappa}(j))$

indexed by the strictly order-preserving maps of Theorem 1.1(b)? Which X_κ are Gorenstein, in terms of the interior of $(\mathcal{O}(P_\kappa), \Lambda)$?

Conjecture 8.3 (A half-step operator)**.** *Let T be the unipotent square root of $A_o A_e$ (the matrix of (2)), indexed by strict partitions ordered by inclusion. Then:*

(a) $s_{x+1/2} = T\, s_x$ for every x; equivalently, $T\, s_0 = s_{1/2}$;

(b) each entry $T(\kappa, \nu)$ is 0 or $\pm 2^{-k}$. It is multiplicative over the connected components of $\theta = D(\kappa) \setminus D(\nu)$. An off-diagonal single box has weight 1; an off-diagonal component with at least two boxes has weight 0; and a component containing exactly one diagonal box is a ribbon D of weight $\pm 2^{-|D|}$, with sign $+$ iff the left neighbour of its north-east end box lies in D.

Part (b) was checked on all 1716 pairs at $m \le 6$, and part (a) for $m \le 4$. Neither part is used in this paper. A natural guess, that T is given by a shifted Murnaghan–Nakayama rule, is false: the two agree only on single boxes, and their supports differ.

Question 8.4 (Orthogonal half-integers)**.** On $\mathrm{OG}(m, 2m+1)$ the half-integer values of H^{OG}_κ have sporadic zeros, for example $H^{\mathrm{OG}}_{(3,2)}(-\frac{5}{2}) = 0$ and $H^{\mathrm{OG}}_{(5,1)}(-\frac{3}{2}) = 0$. We found ten such κ at $m = 6$, and no pattern. Is there an OG analogue of Theorem 1.5?

Question 8.5 (Semi-toric degenerations)**.** Chirivì's semi-toric degeneration of X_κ [5], as summarised in [6, §7], has one toric component for each maximal chain $\varnothing < \cdots < \kappa$, that is, for each standard shifted tableau of shape κ. The bond vector of such a chain lies in $\{1, 2\}^{|\kappa|}$, with 1 exactly at diagonal boxes. The associated lecture hall polytope appears to be the simplex of Stanley's canonical triangulation of $\mathcal{O}(P_\kappa)$ that corresponds to the chain, taken in the lattice Λ. Is Chirivì's degeneration a geometric realisation of this triangulation of $(\mathcal{O}(P_\kappa), \Lambda)$? We have not compared it with Chirivì's gluing data.

Question 8.6 (Higher Pieri coefficients)**.** Proposition 4.6 is the case $p = 1$ of the KLG-tableau formula [4, Cor. 5.6], whose proof is left to the reader there. A complete proof for all p, by the same induction over Theorem 5.5 of [4], would be useful. Do the coefficients $c^{\nu}_{\lambda,p}$ admit an interpretation by strict maps into a $2p$-letter alphabet $\{e_1 < o_1 < \cdots < e_p < o_p\}$ with even letters on the diagonal, extending Theorem 4.7?

Acknowledgements

Clio Vega refereed the core arguments of this paper. She endorsed the hyperplane-section proof of Theorem 6.7 (review uid451), at which point the Chevalley input (G0) was made explicit at her request. She endorsed the inductive proof of Theorem 1.4 for LG and OG (review uid457), found the missing hypothesis $t \le r$ in Lemma 5.2(i), and supplied the short proof of Lemma 5.2(ii) from Theorem 4.7. She also replaced our interpolation-based checks at $m = 5$, which had not finished, by a direct evaluation of Demazure characters at negative weights, extending the verification to $m \le 6$. This draft was prepared for Robin Langer's publication queue.

References

1. H. H. Andersen, *Schubert varieties and Demazure's character formula*, Invent. Math. **79** (1985), 611–618.
2. Matthias Beck and Sinai Robins, *Computing the continuous discretely*, 2nd ed., Undergraduate Texts in Mathematics, Springer, New York, 2015.
3. Michel Brion and Shrawan Kumar, *Frobenius splitting methods in geometric representation theory*, Progress in Mathematics, vol. 231, Birkhäuser, Boston, 2005.
4. Anders Skovsted Buch and Vijay Ravikumar, *Pieri rules for the K-theory of cominuscule Grassmannians*, J. Reine Angew. Math. **668** (2012), 109–132, arXiv:1005.2605. Numbering in this paper follows the arXiv source.
5. Rocco Chirivì, *LS algebras and application to Schubert varieties*, Transform. Groups **5** (2000), 245–264.

6. Rocco Chirivì, Andrea Costa Cesari, Xin Fang, and Peter Littelmann, *Lecture hall polytopes and Lakshmibai–Seshadri paths*, 2026, arXiv:2609.03195.
7. Corrado De Concini, *Symplectic standard tableaux*, Adv. Math. **34** (1979), 1–27.
8. R. Dehy, *Combinatorial results on Demazure modules*, J. Algebra **205** (1998), 505–524.
9. Michel Demazure, *Désingularisation des variétés de Schubert généralisées*, Ann. Sci. École Norm. Sup. (4) **7** (1974), 53–88.
10. N. Gonciulea and V. Lakshmibai, *Degenerations of flag and Schubert varieties to toric varieties*, Transform. Groups **1** (1996), 215–248.
11. Takayuki Hibi, *Distributive lattices, affine semigroup rings and algebras with straightening laws*, Commutative Algebra and Combinatorics (Kyoto, 1985), Adv. Stud. Pure Math., vol. 11, North-Holland, Amsterdam, 1987, pp. 93–109.
12. Anthony Joseph, *On the Demazure character formula*, Ann. Sci. École Norm. Sup. (4) **18** (1985), 389–419.
13. Steven L. Kleiman, *Toward a numerical theory of ampleness*, Ann. of Math. (2) **84** (1966), 293–344.
14. Shrawan Kumar, *Demazure character formula in arbitrary Kac–Moody setting*, Invent. Math. **89** (1987), 395–423.
15. V. Lakshmibai and C. S. Seshadri, *Geometry of G/P. V*, J. Algebra **100** (1986), 462–557.
16. Peter Littelmann, *A Littlewood–Richardson rule for symmetrizable Kac–Moody algebras*, Invent. Math. **116** (1994), 329–346.
17. ———, *The path model for representations of symmetrizable Kac–Moody algebras*, Proceedings of the International Congress of Mathematicians, Zürich, 1994, vol. 1, Birkhäuser, Basel, 1995, pp. 298–308.
18. I. G. Macdonald, *Polynomials associated with finite cell-complexes*, J. London Math. Soc. (2) **4** (1971), 181–192.
19. Olivier Mathieu, *Formules de caractères pour les algèbres de Kac–Moody générales*, Astérisque **159–160** (1988).
20. ———, *Le modèle des chemins*, Séminaire Bourbaki, Vol. 1994/95, Exp. No. 798, Astérisque, vol. 237, 1996, **[CHECK: title and page range not verified; §4 was read from a page image]**, pp. 209–224.
21. Robert A. Proctor, *Bruhat lattices, plane partition generating functions, and minuscule representations*, European J. Combin. **5** (1984), 331–350.
22. A. Ramanathan, *Schubert varieties are arithmetically Cohen–Macaulay*, Invent. Math. **80** (1985), 283–294.
23. C. S. Seshadri, *Introduction to the theory of standard monomials*, Texts and Readings in Mathematics, vol. 46, Hindustan Book Agency, New Delhi, 2007.
24. Richard P. Stanley, *A chromatic-like polynomial for ordered sets*, Proc. Second Chapel Hill Conference on Combinatorial Mathematics and its Applications, 1970, pp. 421–427.
25. ———, *Two poset polytopes*, Discrete Comput. Geom. **1** (1986), 9–23.
26. ———, *Enumerative combinatorics, volume 1*, 2nd ed., Cambridge Studies in Advanced Mathematics, vol. 49, Cambridge University Press, 2012.
27. John R. Stembridge, *Enriched P-partitions*, Trans. Amer. Math. Soc. **349** (1997), 763–788.